

An $O(n^{1/6})$ -Coloring Algorithm for 3-Colorable Graphs

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Abstract

We study polynomial-time coloring of 3-colorable n -vertex graphs. Williamson and Shmoys list finding a polynomial-time $O(\log n)$ -coloring algorithm for such graphs among the ten open problems in *The Design of Approximation Algorithms* [WS11]. The recent 4-to-1 games result of Fei, Minzer, and Wang [FMW26] implies that coloring such graphs with any fixed constant number of colors is NP-hard. The previous best upper bound, due to Bansal, Huang, and Lee, is $O(n^{0.19539})$ [BHL26]. We give a randomized polynomial-time algorithm using $O(n^{0.16523})$ colors.

*magic.linuxkde@gmail.com. The author started working on this problem from August 2026. The author had planned to release the draft only once the result was tight and could not be improved further. However, due to recent progress on unique games, the author decided to release this draft now. The author would like to thank Victor Reis for mentioning this problem during the author's visit to Microsoft Research Redmond. The author would like to thank Victor Reis, and Lichen Zhang for very helpful discussions.

1 Introduction

Given an n -vertex graph that is promised to be 3-colorable, how few colors can a polynomial-time algorithm guarantee? Very little about this question is settled. Earlier work established NP-hardness of coloring a 3-colorable graph with four colors [GK04] and then five colors [BBKO21], while conditional results ruled out every constant number of colors under variants of the Unique Games Conjecture [DMR09]. On the algorithmic side no bound of the form $n^{o(1)}$ is known. The history of the problem is therefore the history of a single exponent, and it has moved slowly.

Wigderson’s trick opened the line with $O(n^{1/2})$ colors [Wig83], and Blum improved it to $\tilde{O}(n^{3/8})$ by combinatorial means [Blu94]. Karger, Motwani, and Sudan then introduced the vector-coloring relaxation together with its hyperplane rounding and obtained $\tilde{O}(n^{1/4})$ [KMS98]. That rounding, and the local parameter it attaches to a family of vectors with bounded pairwise inner products, is still the engine behind every semidefinite bound, including the one proved here. Blum and Karger combined the semidefinite and combinatorial approaches for $\tilde{O}(n^{3/14}) = \tilde{O}(n^{0.2143})$ [BK97]. Arora, Chlamtác, and Charikar [ACC06], and then Chlamtác [Chl07], brought in the Arora–Rao–Vazirani algorithm for sparsest cut and the sum-of-squares hierarchy, reaching $\tilde{O}(n^{0.2111})$ and $\tilde{O}(n^{0.2072})$. Their mechanism is the one analyzed in this paper: under the assumption that the algorithm of [KMS98] fails globally, a second neighborhood is forced to look random, and a large independent set is extracted from it.

Progress then returned to combinatorial methods, with Kawarabayashi and Thorup reaching $\tilde{O}(n^{0.19996})$ [KT17] and, together with Yoneda, $\tilde{O}(n^{0.19747})$ [KTY24]. Bansal, Huang, and Lee have gone back to the semidefinite side very recently. Extending the analysis of second neighborhoods to third neighborhoods, and supplying a new vector 5/2-coloring, they obtain $O(n^{0.19539})$ [BHL26]. That bound is the previous record and the starting point of this paper.

In a recent work, Fei, Minzer, and Wang [FMW26] prove the 4-to-1 Games Conjecture with perfect completeness. Together with the reduction of Guruswami and Sandeep [GS20], this implies that, for every fixed integer $k \geq 3$, finding a proper k -coloring of a 3-colorable graph is NP-hard. This establishes hardness for every constant number of colors, while leaving a substantial gap to the polynomial-color guarantees considered here.

Williamson and Shmoys [WS11, Chapter 17, Problem 9] include coloring 3-colorable graphs among the ten open problems in their book *The Design of Approximation Algorithms*. They ask for a polynomial-time algorithm that colors every n -vertex 3-colorable graph with $O(\log n)$ colors, presenting this logarithmic bound as an open algorithmic target.

1.1 Our Result

We state our result as follows.

Theorem 1.1 (Main result, informal version of Theorem 4.25). *A randomized polynomial-time algorithm colors every 3-colorable n -vertex graph with $O(n^{0.16523})$ colors.*

Organization of this paper. Section 2 gives an overview of the proof and compares our approach with that of Bansal, Huang, and Lee. Section 3 introduces the notation and recalls the required facts about Gaussian covers, packings, and vector-coloring rounding. Section 4 develops the local rounding procedures and the endpoint-state algorithm, and proves the main result. Section 5 defines the rational threshold profiles and verifies both the local chain recurrences and the profile inequalities. Finally, Section 6 derives the consequences for 5-, 7-, and 9-colorable graphs.

Table 1: Polynomial-time coloring bounds for 3-colorable graphs.

Year	Authors	Refs	Number of colors
1983	Wigderson	[Wig83]	$O(n^{1/2})$
1994	Blum	[Blu94]	$\tilde{O}(n^{3/8})$
1994	Karger, Motwani, and Sudan	[KMS98]	$\tilde{O}(n^{1/4})$
1997	Blum and Karger	[BK97]	$\tilde{O}(n^{3/14})$
2006	Arora, Chlamtác, and Charikar	[ACC06]	$\tilde{O}(n^{0.2111})$
2007	Chlamtác	[Chl07]	$\tilde{O}(n^{0.2072})$
2017	Kawarabayashi and Thorup	[KT17]	$\tilde{O}(n^{0.19996})$
2024	Kawarabayashi, Thorup, and Yoneda	[KTY24]	$\tilde{O}(n^{0.19747})$
2026	Bansal, Huang, and Lee	[BHL26]	$O(n^{0.19539})$
2026	This paper	Theorem 1.1	$O(n^{0.16523})$

2 Technique overview

Section 2.1 reviews the approach of Bansal, Huang, and Lee [BHL26] and identifies the ingredients retained in our proof. Section 2.2 explains our approach and how its new structural and analytic tools extend those ingredients.

2.1 The BHL26 approach

Bansal, Huang, and Lee [BHL26] combine an SDP algorithm for bounded-degree graphs with the dense-graph algorithm of Kawarabayashi, Thorup, and Yoneda. Their sparse/dense reduction [BHL26, Corollary 2.7], reduces the coloring problem to making progress on graphs of maximum degree at most a threshold D . On the sparse side they run the Gaussian KMS rounding algorithm, stated quantitatively in their Theorem 2.3 and in our Lemma 3.4. If this rounding already returns a sufficiently large independent set, the sparse case is finished. The interesting case is therefore the no-progress branch, where most Gaussian-selected vertices have a selected neighbor.

The no-progress branch is converted into geometry through covers and packings. These notions are taken from [BHL26, Definitions 4.3 and 4.5] and reproduced as Definitions 3.1 and 3.3. A maximal matching among the selected vertices assigns mass to normalized edge directions; [BHL26, Example 4.7] shows that this mass is a packing. Because the number of directions is bounded by D , the packing is only mildly inefficient. The spread theorem [BHL26, Theorem 4.11], removes little mass and makes the remaining packing spread. The pruning theorem [BHL26, Theorem 4.12], applies this fact simultaneously along two-step walks, producing a nonempty induced subgraph in which the relevant rooted-neighborhood packings have controlled mass and correlation.

Two Gaussian operations turn the pruned two-step structure into a large second neighborhood. The projection lemma [BHL26, Lemma 4.15], projects a cover away from one anchor while losing only a Gaussian tail. The composition lemma [BHL26, Lemma 4.14], composes an outer inefficient cover with orthogonal fiber covers. Its loss is governed by Gaussian isoperimetry and is of order $\sqrt{c}s_1$ in their parameter regime, where s_1 is the outer cover threshold and c its inefficiency parameter. The composed cover forces many second-neighborhood vertices. At the same time, the pruned correlations put those vertices close to root correlation $1/4$, where the positive-correlation construction [BHL26, Lemma 3.4], recorded as our Lemma 3.6, gives a vector $2 + o(1)$ coloring. KMS rounding then extracts an independent set containing almost a constant fraction of that

second neighborhood.

The new step in BHL26 is to continue to a third neighborhood. The third-neighborhood pruning theorem [BHL26, Theorem 5.8], prunes three-step walks so that a family W of second-level endpoints is itself a cover and each endpoint retains a substantial third-step fiber cover. The composition lemma [BHL26, Lemma 5.9] combines these two layers using [BHL26, Lemmas 4.14 and 4.15]. A win-win argument separates the case in which W is already much larger than its union-bound lower estimate from the case in which W is efficient enough for composition. The typical third-level root correlation is near $-1/8$. The anchor construction [BHL26, Lemma 3.2], recorded as our Lemma 3.5, uses the local distributions of the three-round SoS relaxation to turn this information into their new vector $5/2$ coloring. KMS rounding loses a factor about $D^{1/5}$ on the resulting third neighborhood, but the extra neighborhood growth compensates for that loss. The win-win optimization [BHL26, Theorem 5.10] balances the two branches, [BHL26, Corollary 5.11] supplies the sparse independent set, and [BHL26, Corollary 2.7] combines it with the dense routine to obtain exponent 0.19539.

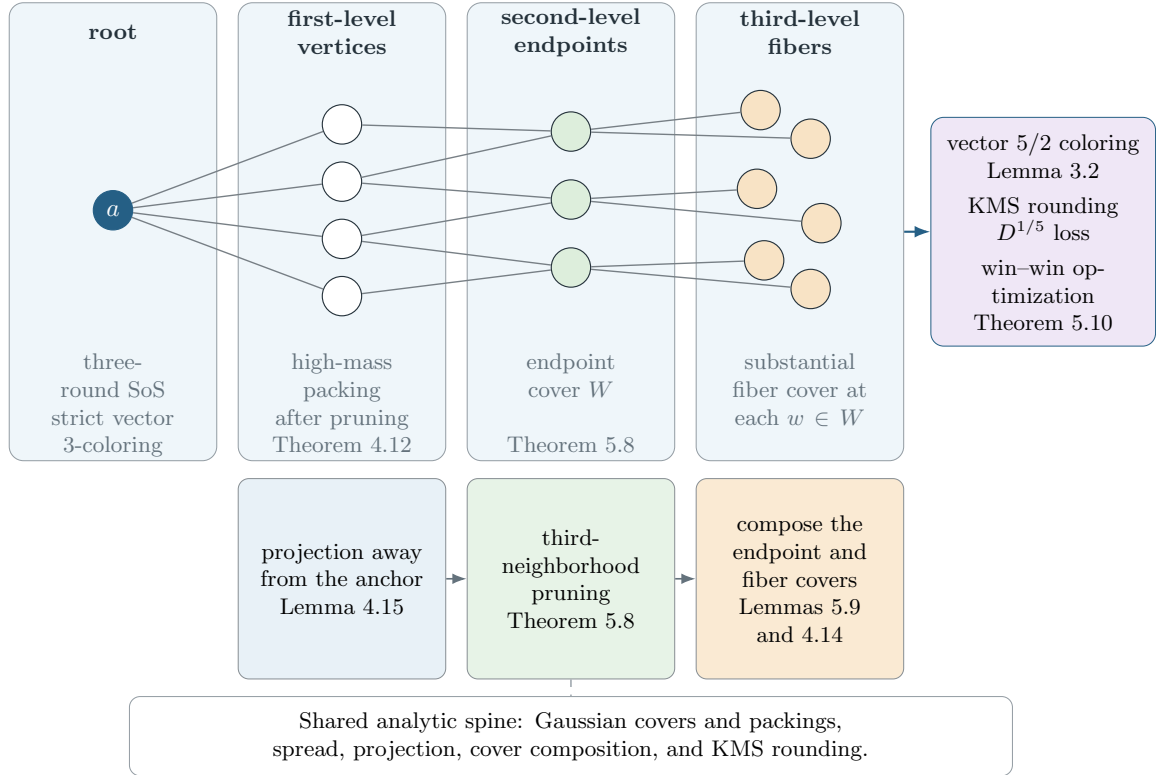


Figure 1: The fixed three-neighborhood pipeline of Bansal, Huang, and Lee [BHL26, Theorem 5.8 and Lemma 5.9]. All displayed theorem and lemma numbers refer to the numbering in [BHL26]. The diagram separates the graph layers from the Gaussian projection and cover-composition operations.

2.2 Our approach and comparison

The common spine. We retain the principal algorithmic and geometric language of BHL26. Both proofs use the sparse/dense reduction [BHL26, Corollary 2.7] (Lemma 4.4), Gaussian KMS rounding [BHL26, Theorem 2.3] (Lemma 3.4), the color-symmetric three-round SoS moment sys-

tem, and Gaussian covers and packings from [BHL26, Definitions 4.3–4.5]. The matching-to-packing implication [BHL26, Example 4.7] is isolated as Lemma 4.6; its robust implemented form is Lemma 4.7. We also use the same broad principle as [BHL26, Lemmas 4.14 and 4.15]: project an outgoing packing and compose it with a cover at the preceding level. Thus the improvement does not replace the SDP, KMS rounding, or the dense-graph algorithm.

Nested matching menus. BHL26 use an explicitly pruned configuration of three neighborhoods. Lemma 4.8 instead constructs a fixed number of nested layers supporting every control in a deterministic $n^{o(1)}$ -sized menu. Its marking argument either produces the target independent set or removes only a sublinear set of vertices. Consequently the control may depend on a parent correlation even though the layers are prepared before the root is selected. Every surviving vertex has every required packing into the preceding layer.

Correlation-sensitive rounding and tilted projection. Lemma 4.5 combines the negative-correlation anchor construction [BHL26, Lemma 3.2] and the positive-correlation construction [BHL26, Lemma 3.4] into the piecewise loss $L^*(s)$. Lemma 4.17 applies five fixed-depth local procedures to each strict stratum. A zero-tilt chain uses KMS rounding on projected strata. A tilted chain uses that zero-tilt procedure, and three further tilted chains use both of these preceding procedures. The last two use the smaller target bound $1247/1000$, and one also tilts the first propagation while keeping seed creation unchanged. Each tilted chain selects its control from the parent correlation. Their dependence is finite, and both child signs are retained in the tilted analysis. This replaces the original loss $L^*(s)$ by the smaller loss $G(L^*(s))$. Failure to obtain verified progress bounds a covered stratum’s cardinality by the cap $M(s)$ of Definition 4.19. For propagation, Lemma 4.10 conditions the Gaussian anchor coordinate near $z(s)t$. This changes the projected threshold to $\sqrt{3q(s)} - z(s)y$ and spends $z(s)^2$ of the squared-tail budget. The two exact reserves in Eq. (9) pay for both simultaneous peeling and projection. Retention uses its explicit quadratic threshold test, rather than a fixed cutoff alone.

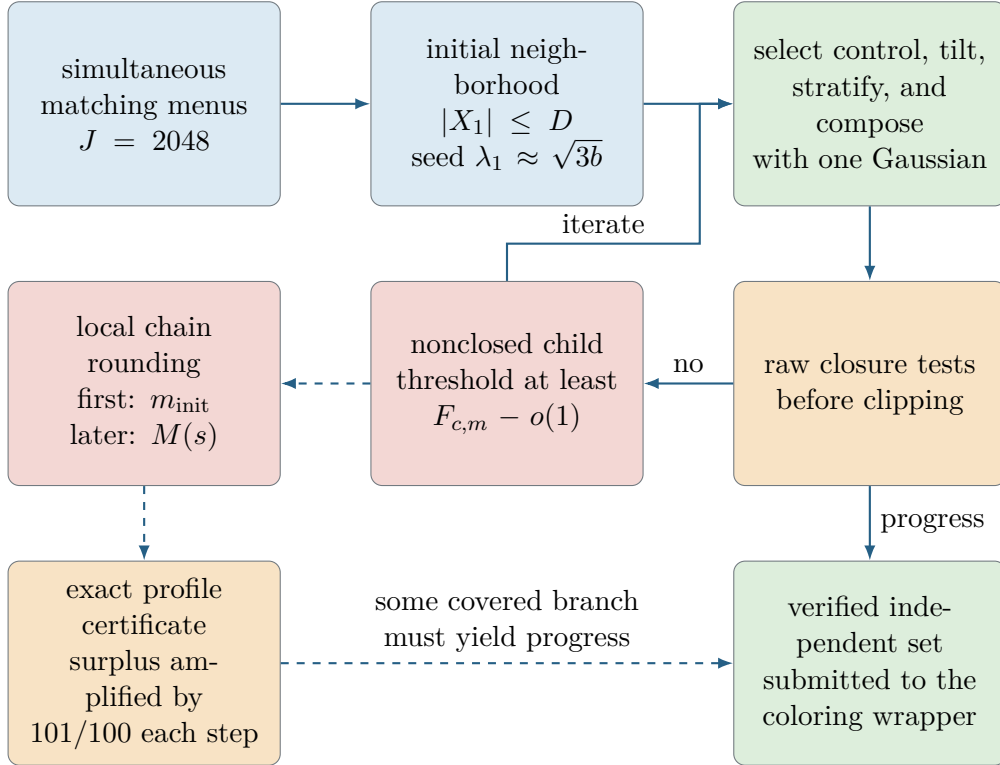
Heterogeneous common-anchor composition. Lemma 4.11 keeps the actual projected norm and base threshold of every retained fiber. A single interpolation coefficient relates three convex simultaneous-failure sets coordinate by coordinate. Ehrhard’s inequality then gives the required fiber failure bound. In the outer union bound, Gaussian independence is used only for each fixed endpoint and its perpendicular fiber; no independence among different endpoints is required. The control is fixed from the parent before a successful child is identified.

Degree-aware initialization and endpoint states. The first family is a root neighborhood, so its cardinality is at most D . Its first transition uses $m_{\text{init}} = \beta + e$ and a special first control. Every later transition uses $m = M(s)$ and the parent-selected profile control. Lemma 4.21 enumerates all retained candidates and computes a certified minimum threshold and witness for each bin. Lemma 4.22 tests raw closure before clipping, so a covered continuing child loses only $o(1)$ when its threshold is stored. An uncovered candidate may instead be discarded without a progress claim.

Profile termination. The scalar certificate defines a continuous piecewise-affine profile U generated by 100 fixed updates from 47 rational control values. Lemma 5.6 proves that the first transition enters above U , and that every later admissible transition amplifies the normalized surplus $\mathcal{P}(s, \lambda) = \lambda/U(s) - 1$ by a factor $101/100$, with a fixed additive reserve. This covers every threshold from $U(s)$ to the cap, including branch hinges and cap endpoints. Exact integer interval

arithmetic checks complete rectangles and hulls all knot and cap seams. Separate exact lower-envelope certificates prove termination of local chains 0, 1, 2, 4, 5 by state depths 9, 28, 37, 40, 25, respectively. The fixed profile is verified by forward inequalities; its construction is not a proof assumption.

Algorithmic bookkeeping. The nonclosure cap bounds the surplus, so Corollary 4.24 forces progress within a fixed $J = 2048$ layers. The profile construction and the certificate grid are fixed proof data; the graph algorithm uses finer, input-dependent bins and a deterministic control menu. Lemma 4.2 supplies rational moment data, adaptive Gaussian coupling, and certified scalar comparisons after the complete polynomial work bound has been fixed. The fixed certificate reserves absorb all vanishing errors. This yields Theorem 4.25 through the inherited sparse/dense wrapper.



Solid arrows describe the graph algorithm;
dashed arrows describe its termination proof.

Figure 2: The parent-controlled threshold-profile method. The graph algorithm enumerates all candidates; the proof follows a covered chain. Initial entry and amplified surplus force verified progress at fixed depth.

3 Preliminaries

This section introduces the notation and recalls the tools from Bansal, Huang, and Lee [BHL26] used in the proof. Section 3.1 fixes the graph, vector, Gaussian, and asymptotic notation. Section 3.2 introduces covers, packings, and inefficiency, and records their basic properties. Section 3.3 gives

the quantitative KMS rounding procedure and the vector-coloring guarantees for nonpositive and positive anchor correlations.

3.1 Notation

All graphs are finite, simple, and undirected. For a graph $G = (V, E)$, put $n := |V|$. For $i \in V$, write $N_G(i) := \{j \in V : ij \in E\}$, $d_G(i) := |N_G(i)|$, and $\Delta(G) := \max_{i \in V} d_G(i)$. For $S \subseteq V$, the notation $G[S]$ denotes the subgraph induced by S . We omit the subscript G when the ambient graph is clear. For a positive integer m , write $[m] := \{1, \dots, m\}$.

Vectors are written in boldface. We use $\mathbf{x} \cdot \mathbf{y}$ for the Euclidean inner product, $\|\mathbf{x}\|$ for the Euclidean norm, and $\mathbf{x} \perp \mathbf{y}$ for orthogonality. Probabilities and expectations are taken over the displayed random choices and are written as $\Pr[\cdot]$ and $\mathbb{E}[\cdot]$, respectively.

A vector κ -coloring of G assigns a unit vector $\mathbf{v}_i$ to each $i \in V$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \leq -1/(\kappa - 1)$ for every $ij \in E$; it is strict if equality holds on every edge.

All logarithms are natural. The symbols $\tilde{O}(\cdot)$ and $\tilde{\Omega}(\cdot)$ suppress factors polylogarithmic in n . Unless stated otherwise, asymptotic notation refers to the limit $n \rightarrow \infty$, and its hidden constants may depend on fixed numerical parameters but not on n .

Throughout, $\mathbf{r}$ denotes a standard Gaussian vector in the relevant Euclidean space. If $Z \sim N(0, 1)$, write $\bar{\Phi}(t) := \Pr[Z \geq t]$ for the standard Gaussian upper tail. We use the color-symmetric three-round SoS/Lasserre relaxation of [BHL26]. Its moment vectors induce a strict vector 3-coloring $\{\mathbf{v}_i\}$. For an oriented edge ij , we write $\mathbf{v}_{ij}$ for the normalized projection of $\mathbf{v}_j$ onto $\mathbf{v}_i^\perp$. We write KMS' for the modified rounding routine of [KMS98] used in [BHL26].

3.2 Covers, packings, and inefficiency

Gaussian covers describe the threshold events that drive neighborhood growth.

Definition 3.1 (Cover and inefficiency, Definition 4.3 of [BHL26]). *A finite set X of unit vectors is an (σ, δ) -cover if*

$$\Pr_{\mathbf{r}}[\exists \mathbf{x} \in X : \mathbf{r} \cdot \mathbf{x} \geq \sigma] \geq \delta.$$

It is at most c -inefficient with respect to σ if $|X| \leq \bar{\Phi}(\sigma)^{-(1+c)}$.

A union bound converts a cover probability into a lower bound on its size.

Fact 3.2 (Cover counting, Fact 4.4 of [BHL26]). *If X is an (σ, δ) -cover, then $|X| \geq \delta \bar{\Phi}(\sigma)^{-1}$.*

Packing measures make these cover bounds available after restricting to a subfamily.

Definition 3.3 (Packing, indexed form of Definition 4.5 of [BHL26]). *Let $(\mathbf{x}_j)_{j \in I}$ be a finite indexed family of unit vectors and let μ be a measure on the index set I . The pair $((\mathbf{x}_j)_{j \in I}, \mu)$ is an (σ, δ) -packing if $\mu(I) \geq \delta$ and*

$$\mu(I') \leq \Pr_{\mathbf{r}}[\exists j \in I' : \mathbf{r} \cdot \mathbf{x}_j \geq \sigma] \quad \text{for every } I' \subseteq I.$$

Repeated geometric vectors are allowed. Equivalently, one may aggregate the weights of equal vectors, recovering the set-valued definition of [BHL26].

Taking $I' = I$ shows that the underlying vector family is an (σ, δ) -cover, and the displayed inequality is inherited by subfamilies. Thus restriction to $I' \subseteq I$ is an $(\sigma, \mu(I'))$ -packing. Running KMS' on G and recording, for each vertex i , the neighborhood vectors together with the measure induced by the canonical maximizer yields a packing $(\{\mathbf{v}_{ij} \mid j \in N(i)\}, \mu_i)$ for every i ; this is Example 4.7 of [BHL26].

3.3 Rounding and vector colorings

We use a quantitative rounding guarantee whose constants remain uniform as the vector-coloring parameter varies.

Lemma 3.4 (Quantitative KMS rounding, Theorem 2.3 of [BHL26]). *There are absolute constants $c_{\text{KMS}} > 0$ and $A_{\text{KMS}} > 0$ with the following property. Let $G = (V, E)$ have at most n vertices, let $2 \leq \kappa \leq 31/10$, suppose that G admits a vector κ -coloring, and let $D \geq \max\{2, \Delta(G)\}$. Put $\rho := (\kappa - 2)/\kappa$. One run of the randomized polynomial-time procedure **QUANTITATIVEKMS** (Algorithm 1), with degree-bound normalization and a deterministic greedy fallback for bounded degree, returns an independent set I satisfying*

$$\mathbb{E}[|I|] \geq c_{\text{KMS}} |V| D^{-\rho} \{\log(n+2)\}^{-A_{\text{KMS}}}.$$

The constants are uniform on the displayed compact range of κ . Consequently the usual $\tilde{\Omega}(|V| D^{-\rho})$ guarantee follows.

Algorithm 1 Quantitative KMS rounding

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1: procedure QUANTITATIVEKMS( $G, (\mathbf{v}_i)_{i \in V}, \kappa, D$ ) ▷ Lemma 3.4
2:    $D \leftarrow \min\{D, \max\{2, |V|\}\}.$ 
3:    $\rho \leftarrow (\kappa - 2)/\kappa$  and  $L \leftarrow \log(D + 2).$ 
4:   if  $D$  is at most the fixed absolute cutoff then
5:     return a maximal independent set obtained greedily.
6:   end if
7:    $q \leftarrow \sqrt{2\rho(\log D + 3 \log L)}.$ 
8:   Sample  $g \sim \mathcal{N}(0, I)$  and set  $S \leftarrow \{i \in V : g \cdot \mathbf{v}_i \geq q\}.$ 
9:    $I \leftarrow \{i \in S : N_G(i) \cap S = \emptyset\}.$ 
10:  return  $I.$ 
11: end procedure

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Proof. We include the one-run calculation in order to make the uniformity explicit. Write D_{in} for the supplied bound. The algorithm sets $D := \min\{D_{\text{in}}, \max\{2, |V|\}\}$ and uses a fixed cutoff at least 2. If $|V| \leq 1$, the greedy branch returns V . Otherwise $\max\{2, \Delta(G)\} \leq D \leq |V| \leq n$. Since $\rho \geq 0$, the estimate for this D implies the one for D_{in} . Put $L := \log(D + 2)$. If $\rho > 0$, run KMS with Gaussian threshold

$$q^2 := 2\rho\{\log D + 3 \log L\}.$$

The one-dimensional Mills bound, uniformly for $q \geq 0$, gives

$$p := \Pr[Z \geq q] \geq c_0(1+q)^{-1} e^{-q^2/2} \geq c_1 D^{-\rho} L^{-3\rho-1/2}. \quad (1)$$

For the Gaussian forms X, Y belonging to the endpoints of an edge, their correlation r is at most $-1/(\kappa - 1)$. If $r = -1$ their joint upper-tail probability is zero. Otherwise the implication $\{X \geq q, Y \geq q\} \subseteq \{X + Y \geq 2q\}$ and the exponential-moment bound for the normal variable $X + Y$ give

$$\Pr[X \geq q, Y \geq q] \leq \exp\left(-\frac{q^2}{1+r}\right) \leq D^{-(1+\rho)} L^{-3(1+\rho)}. \quad (2)$$

Here we used $2\rho(\kappa - 1)/(\kappa - 2) = 2(\kappa - 1)/\kappa = 1 + \rho$. Let I be the selected vertices having no selected neighbor. Summing the union bound at every vertex and then using Eq. (1) and Eq. (2) yields

$$\mathbb{E}[|I|] \geq |V| \{p - D \Pr[X \geq q, Y \geq q]\} \geq \frac{c_1}{2} |V| D^{-\rho} L^{-3\rho-1/2}$$

once D exceeds an absolute constant: the ratio of the second term to the first is at most $CL^{-5/2}$, independently of κ . Bounded D is covered, with a smaller absolute constant, by returning a greedy independent set. When $\rho = 0$, every edge has correlation -1 ; the threshold $q = 0$ selects no two adjacent vertices (up to a null boundary event) and has expectation $|V|/2$. Finally $0 \leq \rho \leq 11/31$ on the stated interval and $L \leq C \log(n + 2)$. Taking, for example, $A_{\text{KMS}} = 2$ and decreasing the absolute prefactor proves the lemma. This is the quantitative form of the calculation underlying Theorem 2.3 of [BHL26]. $\square$

A common anchor improves the vector coloring when all root correlations are nonpositive.

Lemma 3.5 (Anchor vector coloring, Lemma 3.2 of [BHL26]). *Let the strict vector 3-coloring be induced by a color-symmetric three-round SoS/Lasserre solution. Let $s \leq 0$ and let $i \in V$ satisfy $\mathbf{v}_i \cdot \mathbf{v}_j \leq s$ for every $j \in V \setminus \{i\}$. Then the subgraph induced by $V \setminus \{i\}$ admits a vector $\kappa(s)$ -coloring, where*

$$\kappa(s) := \frac{3 - 6s}{1 - 4s}.$$

The hypothesis $s \leq 0$ is used. For $s > 0$ the anchor construction is unavailable, but a second construction of [BHL26] takes over once the correlation is large enough.

Lemma 3.6 (Positive-correlation vector coloring, Lemma 3.4 of [BHL26]). *Let the strict vector 3-coloring be induced by a color-symmetric three-round SoS/Lasserre solution. Let $\frac{1}{16} \leq s \leq \frac{1}{4}$ and let $i \in V$ satisfy $\mathbf{v}_i \cdot \mathbf{v}_j \geq s$ for every $j \in V \setminus \{i\}$. Then the subgraph induced by $V \setminus \{i\}$ admits a vector $\kappa^+(s)$ -coloring, where*

$$\kappa^+(s) := \frac{4 + 8s}{1 + 8s}.$$

In particular $\kappa^+(\frac{1}{4}) = 2$.

Both colorings are read off the same 3-round SoS/Lasserre relaxation, so using them together needs no strengthening of the semidefinite program.

4 A certified bound below the one-sixth exponent

Section 4.1 gives the rounding tools and their finite-precision implementation. Section 4.2 constructs matching menus and nested high-mass layers. Section 4.3 defines the parent-dependent controls and proves the tilted projection estimates. Section 4.4 proves the heterogeneous common-anchor composition bound. Section 4.5 combines these tools into a hierarchy of local rounding procedures. Section 4.6 defines endpoint states and their cardinality caps. Section 4.7 constructs the state transitions and proves survival before clipping. Section 4.8 initializes the states and uses amplified profile surplus to establish termination within a fixed number of layers. Section 4.9 assembles the coloring algorithm and proves the main result.

Put

$$a_0 := \frac{165227499}{10^9} = 0.165227499, \quad \beta := 1 - 2a_0 = \frac{334772501}{500000000}, \quad D := n^\beta.$$

The exact gap $0.1652275 - a_0 = 10^{-9}$ absorbs subpolynomial and polylogarithmic losses. Only the first endpoint family is bounded by D ; later families use the root-correlation cap. We choose the Gaussian control from the parent correlation before the successful child is identified.

We use physical Gaussian units

$$t := \sqrt{2 \log n}, \quad \varepsilon_n := t^{-1/3}, \quad \zeta_n := n^{-C_0}.$$

Thus $\log D = \beta t^2/2$ exactly. A matching experiment with selection parameter q uses threshold $\sqrt{q}t$, not t . The constants q used below range over a fixed positive compact interval. The degree test is implemented by comparing the integers $d^{500000000}$ and $n^{334772501}$.

Choose the fixed precision exponent C_0 after bounding the complete ideal execution, including all amplification trials and rejection proposals. This bound counts arithmetic operations and comparisons, not the bit work of their implementation. Certified rational bisection for the logarithm and square root gives brackets $t^- \leq t \leq t^+$ of width at most ζ_n in polynomial time. Use analogous brackets for all Gaussian tails, control values, and thresholds. Define $m_n := \lceil t^+ \rceil$ and $\omega_n := 2^{-m_n}$. Correlation core bins have width at most $\omega_n/2$; insert all profile knots and structural breakpoints. A comparison interval touching a boundary is assigned to both adjacent bins. The padded bin B^+ includes these rounding collars and has diameter at most ω_n for large n . In particular,

$$\log(1/\omega_n) = O(t) = o(\varepsilon_n t^2), \quad \sqrt{\omega_n} = o(\varepsilon_n), \quad t^{-1/2} = o(\varepsilon_n).$$

All small-error statements below are uniform after the fixed depth, control menu, and polynomial work bound have been fixed. Small orders n are handled directly after the resulting absolute threshold N_0 is chosen.

4.1 Black boxes and root-correlation rounding

Write $\bar{\Phi}(x) := \Pr[Z \geq x]$ for a standard normal random variable Z . We solve the color-symmetric three-round SoS/Lasserre relaxation. We first describe the exact structural system. In the notation of [BHL26], it supplies moment vectors q_S for consistent partial color assignments S involving at most three vertices. Whenever $S \cup T$ involves at most three vertices, they satisfy

$$\mu(S \cup T) := q_S \cdot q_T = \|q_{S \cup T}\|^2.$$

Here both sides are zero if $S \cup T$ is inconsistent or violates an edge, and $\mu(A)$ denotes the consistently defined local probability of the partial assignment A ; no global distribution is assumed. If $S \cup \{(i, C)\}$ involves at most three vertices, marginalization gives

$$\sum_C q_{S \cup \{(i, C)\}} = q_S.$$

All inner products are invariant under a common permutation of the three colors. In particular,

$$q_{\{(i, R)\}} = \frac{1}{3}q_\emptyset + \frac{\sqrt{2}}{3}v_i, \quad v_i \cdot v_j = -\frac{1}{2} \quad (ij \in E).$$

For every pair of vertices a, x , the union in the following identities involves only those two vertices, so consistency and positivity give

$$\|q_{\{(a, R), (x, R)\}}\|^2 = q_{\{(a, R)\}} \cdot q_{\{(x, R)\}} = \frac{1 + 2v_a \cdot v_x}{9} \geq 0.$$

Consequently every actual root-to-endpoint correlation is at least $-1/2$. This pointwise fact remains valid under adaptive exploration because it holds for every vertex pair; it is not asserted for projected directions. Only unions involving at most the three vertices a, u, v occur below, so three rounds suffice.

Three-vertex consistency also restricts the two root correlations across an edge.

Lemma 4.1 (Root-edge triangle inequality). *Let a, x, z be vertices with $xz \in E$, and put $s := v_a \cdot v_x$ and $\tau := v_a \cdot v_z$. Then*

$$s \geq -\frac{1}{2}, \quad \tau \geq -\frac{1}{2}, \quad s + \tau \leq \frac{1}{2}.$$

Proof. The first two inequalities are the pair-moment bound above. For the last one, use the nonnegative local distribution on the three vertices $\{a, x, z\}$. Color symmetry and marginalization give

$$\mu(a = R, x = R) = \frac{1 + 2s}{9}, \quad \mu(a = R, z = R) = \frac{1 + 2\tau}{9}.$$

Because xz is an edge, these two events are disjoint. Both lie in $\{a = R\}$, whose local mass is $1/3$. Therefore

$$\frac{1 + 2s}{9} + \frac{1 + 2\tau}{9} \leq \frac{1}{3},$$

which is equivalent to $s + \tau \leq 1/2$. $\square$

We transfer the moment and Gaussian calculations to rational data and random-bit sampling.

Lemma 4.2 (Ordinary-bit implementation). *Fix constants $A, R > 0$ and an ideal graph execution with deterministic caps allowing at most n^A arithmetic operations, fresh Gaussian proposals, and scalar comparisons in total. These caps include all amplification trials, rejected proposals, and restarts, and are fixed before the numerical precision. There is an effective constant A_{fp} , independent of C_0 , such that for sufficiently large fixed C_0 , with*

$$\rho_n := n^{A_{\text{fp}}} \sqrt{\zeta_n},$$

the execution has a randomized polynomial-time ordinary-bit implementation. Its bit-work exponent may depend on C_0 and is not required to be A . Its rational moment Gram matrix is exactly positive semidefinite and has residual at most ζ_n . All derived vector and scalar quantities admit the $O(\rho_n)$ and $O(\sqrt{\omega_n} + \rho_n)$ enclosures stated in the proof. Except with probability n^{-R} , its complete adaptive Gaussian and conditioned-matching transcript agrees with the ideal transcript on every comparison whose margin exceeds $4\rho_n$. Consequently all cover, matching, and coloring conclusions below transfer to the ordinary-bit model with only these displayed losses.

Proof. We prove the Gram, normalization, and transcript guarantees in this order, choosing C_0 after the complete work and failure budgets.

Rational moment data and stability. Write the affine moment constraints as $\mathcal{A}_n(X) = \mathbf{b}_n$, where X is a symmetric $d_n \times d_n$ matrix. The dimension and coefficient bounds are polynomial in n . Choose a rational $L_n \geq 1$ bounding the sum of absolute coefficients in every constraint, so that

$$\|\mathcal{A}_n(E)\|_\infty \leq L_n \max_{i,j} |E_{ij}|, \quad \xi_n := \frac{\zeta_n}{4L_n}.$$

Minimize $f_n(X) := \|\mathcal{A}_n(X) - \mathbf{b}_n\|_2^2$ over $K_n := \{X \succeq 0 : \text{tr } X \leq 2d_n\}$ by weak optimization and separation. A proper 3-coloring supplies a zero-residual point with trace at most d_n . The trace

ball has polynomial inner and outer radii; no interior point of the exact affine feasible set is needed. Let $H_n \geq 1$ be a polynomial Lipschitz bound for f_n on the unit neighborhood of K_n . Use objective accuracy $\zeta_n^2/64$ and weak-membership and arithmetic tolerances small enough that their combined distance error is at most $\min\{\xi_n, \zeta_n^2/(64H_n)\}$. Thus the rational output X satisfies

$$\|\mathcal{A}_n(X) - \mathbf{b}_n\|_\infty \leq \zeta_n/4, \quad \lambda_{\min}(X) \geq -\xi_n.$$

All requested accuracies have polynomial bit length.

Set $X^+ := X + \xi_n I$, which is positive semidefinite. Approximate its positive-semidefinite square root by a rational matrix B with $\max_{i,j} |(BB^\top - X^+)_{ij}| \leq \xi_n$. The coefficient bound gives

$$\|\mathcal{A}_n(BB^\top) - \mathbf{b}_n\|_\infty \leq \zeta_n/4 + L_n \xi_n + L_n \xi_n = 3\zeta_n/4 < \zeta_n.$$

Thus the vectors $\hat{q}_S$ are rational and their Gram matrix is exactly positive semidefinite, with the claimed affine residual. The squared norm of every vector identity derived from the moment constraints is a fixed linear combination of Gram entries. Its norm is therefore at most $n^{A_{\text{fp}}} \sqrt{\zeta_n}$. The denominators in the edge and state projections below are bounded away from zero on the certified domain. The inequality

$$|\sqrt{x_+} - \sqrt{y_+}| \leq \sqrt{|x - y|}$$

controls their square roots. More specifically, a construction made from $O(1)$ moment vectors with uniformly bounded coefficients has every squared norm and edge inner product within $O(\rho_n)$ of the formula obtained by imposing the moment identities exactly. If its squared norms are at least ν^2 , normalization changes an edge inner product by at most $C_\nu \rho_n$. Decrease the target reciprocal edge parameter from $1/(\kappa - 1)$ to $1/(\kappa - 1) - C_\nu \rho_n$. On $2 \leq \kappa \leq 3$ the corresponding vector chromatic number is $\kappa + O(\rho_n)$. In particular, the implemented edge inner products satisfy

$$\hat{v}_x \cdot \hat{v}_z \leq -\frac{1}{\kappa - 1} + C_\nu \rho_n. \quad (3)$$

This is a direct perturbation of the computed Gram identities, not an assertion that $\hat{q}$ is close to a particular exact feasible solution. The square-root modulus and ordinary outward interval propagation prove the remaining scalar enclosure claim, with width at most $C\{\sqrt{\omega_n} + \rho_n\}$ on every aligned bin.

For an edge xz , put $\hat{s} := \hat{v}_a \cdot \hat{v}_x$ and $\hat{\tau} := \hat{v}_a \cdot \hat{v}_z$. We claim

$$\hat{s} + \hat{\tau} \leq \frac{1}{2} + C_\Delta \rho_n. \quad (4)$$

Expand the two approximate pair marginals through the nonnegative squared norms of the three-vertex assignment vectors. The terms in which both x and z receive color R have total mass $O(\zeta_n)$ by the edge constraints. All remaining terms are disjoint subevents of $a = R$, whose marginal is $1/3 + O(\zeta_n)$. The approximate pair identities convert their sum to $(2 + 2\hat{s} + 2\hat{\tau})/9 + O(\rho_n)$. Rearranging, and using $\zeta_n \leq \rho_n$, proves the displayed collar without assuming that the implemented moments are close to a particular exact feasible solution.

Random-bit Gaussian and conditional transcripts. Fix a constant Q larger than every selection exponent in a conditional sampler, so its ideal acceptance probability is at least n^{-Q} for large n . Before choosing A or C_0 , cap each such sampler at $\lceil n^{Q+1} \rceil$ proposals and fix all amplification and restart budgets. Exhausting a cap returns FAILURE. Enlarge A to count all these

proposals and their comparison batches. The ideal probability of exhausting any one rejection cap is at most $\exp(-n)$.

For the random-bit implementation, let $m \leq n^{A_m}$ be the dimension of the rational Gram representation and expose all m coordinates of a fresh Gaussian independently. Choose an integer $R_n = O(\sqrt{\log n})$ so that, over every coordinate of all at most n^A draws, the probability that a coordinate leaves $[-R_n, R_n]$ is below $n^{-R-2A-Q-20}/2$. Partition this interval into dyadic cells of mesh at most $\rho_n/(8\sqrt{m})$ and use the cell midpoint. Let $\tilde{g} := Q_n(g)$ be the quantized continuous Gaussian: each coordinate is replaced by its cell midpoint, and a tail coordinate gives an explicit failure atom. Approximate every cell mass from below by a certified rational number and round downward to a common dyadic denominator. Choose the denominator and mass accuracies so that the sum of all deficits, over all coordinates of all capped proposals, is below $n^{-R-2A-Q-20}/2$. Assign the remaining mass to the failure atom. This defines the implemented discrete sample $\hat{g}$, drawn exactly from fixed blocks of unbiased bits. A failure atom returns FAILURE. The total-variation comparison is between $\tilde{g}$ and $\hat{g}$, which have the same discrete support; it is bounded by the sum of the mass deficits. We compare g with $\tilde{g}$ in Euclidean distance, not in total variation. Couple the two discrete samples maximally, then use the quantization coupling to the continuous sample. Outside the union of tail and coupling failures, simultaneously for every proposal,

$$\|g - \hat{g}\|_2 \leq \rho_n/8.$$

The number of cells and the dimension are polynomial in n , with exponents that may depend on C_0 . The logarithms of the dyadic denominators are polynomial as well. Constructing and sampling these mass tables therefore has polynomial bit cost, determined after C_0 ; this cost is not included in the previously fixed ideal-work exponent A .

For a vector u fixed before the draw and a fixed threshold c , Gaussian anti-concentration gives

$$\Pr[|g \cdot u - c| \leq 4\rho_n] \leq C\rho_n.$$

All Gaussian tests below use vectors whose norms lie in a fixed compact subinterval of $(0, \infty)$, so normalization and rational vector error change only the constant. Condition on the complete past. The batch of forms queried from the next fresh Gaussian is fixed at that time; the algorithm may evaluate the whole batch before making a matching or branch decision. Sum the preceding anti-concentration estimate over all at most n^A comparisons in all batches. If no tested form lies in its boundary collar and the product coupling succeeds, all threshold sets, fixed deterministic maximal matchings, and KMS outputs agree. Sequential maximal coupling conditional on each realized past therefore bounds the total-variation distance of the entire adaptive transcript, rather than the probability difference of each transcript separately. Lowering an analytic cover threshold by the same collar proves the stated cover transfer.

For the conditional sampler, repeatedly run the preceding coordinatewise rational sampler and accept when the certified comparison $\hat{g} \cdot \hat{v}_i \geq \sqrt{qt}$ succeeds; discard and resample if its interval meets the acceptance boundary. The ideal acceptance probability is

$$p_n = \bar{\Phi}(\sqrt{qt}) = n^{-q+o(1)},$$

uniformly over the fixed compact range of selection exponents q , up to the certified $O(\rho_n)$ threshold collar. If b_n is the one-proposal probability that either the product coupling fails or any Gaussian comparison in the proposal's fixed batch meets its boundary collar, the common-proposal rejection coupling fails before its first accepted proposal with probability at most b_n/p_n ; this is the geometric sum of the bad-proposal probabilities. All vertex-threshold comparisons used by the deterministic

matching are included in the fixed batch for the accepted proposal. Thus the same calculation couples the complete matching transcript, not merely its individual atoms. The expected ideal proposal count is $p_n^{-1} = n^{O(1)}$. The preset cap has ideal failure probability at most $\exp(-n)$; the coupled implementation has only the additional bad-proposal probability just bounded. Thus no new repetition budget depends on C_0 .

Finally choose C_0 so large that

$$\rho_n \leq \min\{\omega_n^2, n^{-R-3A-A_m-Q-40}, \varepsilon_n/10^6\}$$

for all sufficiently large n . This is possible because ρ_n is a fixed negative power of n , whereas $\omega_n = \exp\{-O(\sqrt{\log n})\}$. These choices make every unconditional batch error, every ratio b_n/p_n , and the sum over all adaptive invocations smaller than its displayed budget. The sequential coupling argument proves the lemma. $\square$

The unhatted q_S, v_i above denote the exact structural system and are used only when a displayed moment or edge identity is asserted with equality. The ordinary-bit algorithm uses $\hat{q}_S, \hat{v}_i$. In the algorithmic discussion below, an “exact scalar” means the real value of an expression formed from these implemented vectors before outward interval rounding; it does not mean the corresponding scalar of an unknown exact SDP solution. Whenever the analytic proof displays an exact moment formula, the implemented proof uses its residual version from Lemma 4.2 and charges the resulting error to the stated collar. We retain unhatted notation in the analytic Gaussian calculation to keep the formulas readable.

The preceding lemma is instantiated only after fixing the depth, the desired global failure exponent, all repetition and rejection caps, and a polynomial n^A dominating the ideal arithmetic operations, proposals, and comparisons. The bit-work exponent is determined only after C_0 and is not fed back into this choice of A . Let C_{bin} bound the fixed error transfer from a Gram entry to a directly computed correlation or fiber scalar. Choose $C_0 = C_0(A, R)$ and then N_0 so that, for $n \geq N_0$,

$$\rho_n + \sqrt{\omega_n} \leq \frac{\varepsilon_n}{100}, \quad \delta_{\text{bin}} := C_{\text{bin}}\rho_n \leq \frac{\omega_n}{8},$$

and so that the total threshold loss in each stored transition is below 10^{-12} . We also make the total error in each normalized-surplus comparison smaller than $1/(4 \cdot 10^9)$. These choices include the propagation, witness-to-representative, and downward-clipping errors justified below. The depth is fixed before these accuracy choices. Every $O(\varepsilon_n)$ below denotes the explicit bound obtained from this lemma, enlarged by one fixed universal factor.

All deterministic quantities are stored as certified rational intervals. A proof action justified by $f \geq 0$ is taken only when the lower endpoint of the interval for f is nonnegative. A discard justified by $f < 0$ is made only when its upper endpoint is negative. Every candidate branch runs the amplified KMS routine before it can continue. An interval-straddling closure comparison is retained as a continuing candidate, and its interval radius is charged to the explicit survival collar. On the global event $\mathcal{G}$, failure of the KMS routine to return the target set certifies that its exact sufficient cardinality condition did not hold. If a correlation interval intersects more than one bin, the candidate is placed in every intersected bin. If the certified interval for the retention residual meets $(-\infty, 0]$, the direction is retained. Thus deterministic rounding never deletes a candidate required by the analysis. Since $\rho_n = o(\omega_n)$, only a constant number of copies is created at each comparison. Gaussian-dependent comparisons are covered by Lemma 4.2; their exceptional boundary events are charged to the global failure event.

For a core bin B write

$$B^+ := \{u : \text{dist}(u, \overline{B}) \leq \delta_{\text{bin}}\}.$$

A computed correlation or fiber scalar is assigned to B precisely when its certified interval meets $\overline{B}$. Its exact scalar then lies in B^+ . If two exact scalars are assigned to the same bin, their distance is at most $\omega_n/2 + 2\delta_{\text{bin}} \leq \omega_n$. Every breakpoint of L^* is a core-bin endpoint, and B^+ is split again at any breakpoint it meets. This creates only a constant-factor number of candidate copies.

For every oriented edge ij , define

$$e_{ij} := \frac{2v_j + v_i}{\sqrt{3}}.$$

Then

$$v_j = -\frac{1}{2}v_i + \frac{\sqrt{3}}{2}e_{ij}, \quad e_{ij} \perp v_i, \quad \|e_{ij}\| = 1.$$

Conditional amplification turns expected progress into reliable output across adaptive calls.

Lemma 4.3 (Conditional amplification). *Condition on a history $\mathcal{H}$. Suppose $0 \leq X \leq N$, $0 \leq T < N$, and $\mathbb{E}[X \mid \mathcal{H}] \geq T + \Delta$, where $N, \Delta > 0$ and T are fixed given $\mathcal{H}$. Then*

$$\Pr[X \geq T \mid \mathcal{H}] \geq \frac{\Delta}{N - T}.$$

More generally, if each fresh trial has conditional success probability at least $p_0 > 0$, then r trials fail with probability at most $\exp(-rp_0)$. For at most n^A adaptive calls, each with an efficiently verifiable output and this same conditional lower bound, taking $r \geq p_0^{-1}(A + C + 1) \log n$ per call bounds total failure by n^{-C} .

Proof. Write $p := \Pr[X \geq T \mid \mathcal{H}]$. Then $\mathbb{E}[X \mid \mathcal{H}] \leq T(1 - p) + Np$, which proves the first assertion. Iterated conditioning bounds failure in r trials by $(1 - p_0)^r \leq \exp(-rp_0)$. Condition on the history before each call and sum its failure bound over the at most n^A calls. No independence between the entries contributing to X is needed. $\square$

We use the sparse/dense progress theorem in the following standard form, which follows from the dense-graph algorithm of Kawarabayashi–Thorup–Yoneda and the usual degree decomposition; see [KTY24, BHL26].

Lemma 4.4 (Sparse/dense progress). *Fix $0 < a < 1/4$ and put $\beta := 1 - 2a$. Suppose that on every 3-colorable n -vertex graph of maximum degree at most n^β a randomized polynomial-time algorithm finds an independent set of size $n^{1-a-o(1)}$ with success probability at least n^{-q} for some fixed q , uniformly over every valid input and every transcript fixed before its fresh random seed is drawn. The output is one-sided: independence and the claimed cardinality are deterministically checked before it is accepted. We invoke the polynomial-time dense-progress routine of Kawarabayashi–Thorup–Yoneda in the form used by Bansal–Huang–Lee. Thus a dense call contributes no algorithmic failure probability; its output is a concrete progress object consumed by the deterministic degree-decomposition wrapper. Then, for every fixed $a' > a$, every 3-colorable n -vertex graph can be colored in randomized polynomial time with $O(n^{a'})$ colors.*

Proof. If $a' \geq 1$, giving each vertex its own color already proves the claim. Hence assume $a < a' < 1$. Fix $\rho := (a' - a)/2 > 0$. For all sufficiently large n , the independent set in the hypothesis is Type 1 progress toward an $O(n^{a+\rho})$ -coloring. Since $a < 1/4$, we have $\beta = 1 - 2a > 1/2$. On a graph of minimum degree at least n^β , the Kawarabayashi–Thorup–Yoneda dense-graph theorem makes progress toward

$$n^{o(1)} \sqrt{n/n^\beta} = n^{a+o(1)}$$

colors, and hence toward $O(n^{a+\rho})$ colors for all sufficiently large n . Apply Theorem 2.6 of [BHL26] with the fixed exponent $\alpha := a + \rho$ to the maximum-degree and minimum-degree cases. It gives a $\tilde{O}(n^{a+\rho})$ -coloring, which is $O(n^{a'})$ because $a + \rho < a'$.

For any fixed $R > 0$, apply Lemma 4.3 inside the degree-decomposition wrapper, with $p_0 := n^{-q}$ and at most $n^{A_{\text{sd}}}$ adaptive calls. The sparse hypothesis supplies the conditional success bound and efficient verification. Thus $O(n^q(A_{\text{sd}} + R + 2) \log n)$ fresh repetitions per sparse call give total failure at most n^{-R} ; the deterministic dense calls need no repetition. This remains polynomial. Every recursive application uses its current order m and threshold m^β for $m \geq N_{\text{sd}}$. The finitely many smaller orders are colored directly and absorbed into the absolute constant. Hence the wrapper applies uniformly to every induced subinstance. $\square$

We combine the anchor constructions with KMS rounding to control the loss uniformly on each root-correlation bin.

Lemma 4.5 (Uniform root-correlation rounding). *Let G have maximum degree at most D , fix a vertex a , and let $U \subseteq V(G) \setminus \{a\}$ consist of vertices assigned to a core bin B . Thus every exact root correlation lies in B^+ , whose diameter is at most ω_n . Assume that B^+ has been split so that it does not cross any of $-1/2, 0, 1/4$. For every fixed $C > 0$, a polynomial number of repetitions finds, with probability at least $1 - n^{-C}$, an independent subset of U of size*

$$|U| D^{-\sup_{s \in B^+} L^*(s) - \varepsilon_n},$$

where

$$L^*(s) := \begin{cases} 0, & s \leq -1/2, \\ \frac{1+2s}{3-6s}, & -1/2 \leq s \leq 0, \\ \frac{1-4s}{3}, & 0 \leq s \leq 1/4, \\ 0, & s \geq 1/4. \end{cases}$$

The same statement holds for a point stratum at a breakpoint. Unless $G[U]$ is independent, the construction supplies unit vectors v'_u satisfying, uniformly for $s \in B^+$ and $uv \in E(G[U])$,

$$v'_u \cdot v'_v = \frac{L^*(s) - 1}{L^*(s) + 1} + O(\omega_n + \rho_n). \quad (5)$$

Their normalization denominators have fixed positive lower bounds.

Proof. Let $\hat{s}_u := \hat{v}_a \cdot \hat{v}_u$ be the exact real correlation formed from the implemented vectors before interval rounding. Thus $\hat{s}_u \in B^+$. We first construct a genuine vector coloring of $G[U]$ from the rational moment vectors and only then invoke KMS. Put $\rho_{\text{KMS}}(\kappa) := (\kappa - 2)/\kappa$.

Suppose first that $B^+ \subseteq (-\infty, 0]$. Positivity of the quantity $\hat{p}_u$ defined below, together with its residual identity, implies $\hat{s}_u \geq -1/2 - C_{\text{floor}}\rho_n$. Thus a stratum whose upper endpoint is below $-1/2 - C_{\text{floor}}\rho_n$ is empty. In the remaining negative collar put

$$s_B^\# := \max\{\sup B^+, -1/2\}.$$

Define the implemented anchor vectors

$$\hat{r}_u := \hat{q}_{\{(a,R),(u,G)\}} - \hat{q}_{\{(a,R),(u,B)\}}, \quad \hat{p}_u := \|\hat{q}_{\{(a,R),(u,R)\}}\|^2.$$

The residual moment identities give, uniformly for $u \in U$,

$$\widehat{p}_u = \frac{1 + 2\widehat{s}_u}{9} + O(\rho_n), \quad \|\widehat{r}_u\|^2 = \frac{1}{3} - \widehat{p}_u + O(\rho_n) \geq \frac{2}{9} - O(\rho_n).$$

For an edge uv , the same three-vertex marginalization used in the proof of Lemma 3.5 gives

$$\widehat{r}_u \cdot \widehat{r}_v = -\left(\frac{1}{3} - \widehat{p}_u - \widehat{p}_v\right) + O(\rho_n).$$

After normalization, monotonicity in $\widehat{p}_u, \widehat{p}_v \leq (1 + 2s_B^\sharp)/9 + O(\rho_n)$ yields

$$\frac{\widehat{r}_u}{\|\widehat{r}_u\|} \cdot \frac{\widehat{r}_v}{\|\widehat{r}_v\|} \leq -\frac{1 - 4s_B^\sharp}{2 - 2s_B^\sharp} + O(\rho_n). \quad (6)$$

The displayed $2/9$ lower bound is the uniform denominator required by Eq. (3). Together with that denominator, Eq. (6) shows that these computed vectors are a genuine vector $(\kappa(s_B^\sharp) + O(\rho_n))$ -coloring, where

$$\kappa(s_B^\sharp) := \frac{3 - 6s_B^\sharp}{1 - 4s_B^\sharp}, \quad \rho_{\text{KMS}}(\kappa(s_B^\sharp)) = \frac{1 + 2s_B^\sharp}{3 - 6s_B^\sharp} = L^*(s_B^\sharp).$$

In particular, an implemented $O(\rho_n)$ collar below $-1/2$ produces a vector $2 + O(\rho_n)$ coloring, never a formal vector chromatic number below 2.

Now suppose $B^+ \subseteq [0, \infty)$. The residual three-vertex marginalization gives, on every edge uv ,

$$\widehat{p}_u + \widehat{p}_v \leq \frac{1}{3} + C_{\text{edge}}\rho_n. \quad (7)$$

If $\inf B^+ > 1/4 + C'_{\text{edge}}\rho_n$, the residual formula for $\widehat{p}$ makes Eq. (7) impossible, and $G[U]$ is independent; return U directly, which already meets the lemma's conclusion. Otherwise put

$$s_B^\sharp := \min\{\inf B^+, 1/4\}.$$

This includes the point strata at 0 and $1/4$ and the implemented collar above $1/4$. Write

$$\widehat{w}_{u,C} := \widehat{q}_{\{(a,R),(u,C)\}}, \quad \widehat{p}_u := \|\widehat{w}_{u,R}\|^2,$$

and use certified division to form

$$\widehat{c}_u := -\frac{1/3 - \widehat{p}_u}{\widehat{p}_u}, \quad \widehat{z}_u := \widehat{c}_u \widehat{w}_{u,R} + \widehat{w}_{u,G} + \widehat{w}_{u,B}.$$

On the non-independent part of this stratum, $1/9 - O(\rho_n) \leq \widehat{p}_u \leq 1/6 + O(\omega_n + \rho_n)$ for every nonisolated vertex, by the residual formula for $\widehat{p}_u$, the diameter of B^+ , and $\inf B^+ \leq 1/4 + C'_{\text{edge}}\rho_n$. Isolated vertices may be assigned an arbitrary unit vector. Thus the coefficients on the nonisolated subgraph are uniformly bounded. The residual two- and three-vertex identities give

$$\|\widehat{z}_u\|^2 = \frac{1/3 - \widehat{p}_u}{3\widehat{p}_u} + O(\rho_n) \geq \frac{1}{3} - O(\rho_n), \quad \widehat{z}_u \cdot \widehat{z}_v = -\frac{1}{3} + O(\rho_n) \quad (uv \in E).$$

Thus the positive construction also has a uniform normalization denominator. Normalization, Eq. (7), and monotonicity give

$$\frac{\widehat{z}_u}{\|\widehat{z}_u\|} \cdot \frac{\widehat{z}_v}{\|\widehat{z}_v\|} \leq -\frac{1 + 2s_B^\sharp}{2 - 2s_B^\sharp} + O(\omega_n + \rho_n). \quad (8)$$

By Eq. (8), Eq. (3), and the bin-variation bound this is a genuine vector $(3/(1+2s_B^\sharp) + O(\omega_n + \rho_n))$ -coloring, and

$$\rho_{\text{KMS}}(3/(1+2s_B^\sharp)) = \frac{1-4s_B^\sharp}{3} = L^*(s_B^\sharp).$$

The clamped parameter lies in $0 \leq s_B^\sharp \leq 1/4$; in particular no vector chromatic number below 2 and no zero normalization at correlation 1 is invoked.

Thus a stratum more than $C'_{\text{edge}}\rho_n$ above $1/4$ is handled first as edgeless, while the residual collar is clamped to $1/4$ and receives a vector $2+O(\omega_n + \rho_n)$ coloring. A numerical interval touching a breakpoint is duplicated into the adjacent construction and is never discarded by an emptiness or no-edge test.

The same norm and edge-product identities also prove Eq. (5). Since the bin diameter is at most ω_n , the normalized edge values are $-(1-4s)/(2-2s) + O(\omega_n + \rho_n)$ on the negative branch and $-(1+2s)/(2-2s) + O(\omega_n + \rho_n)$ on the positive branch. Substitution of $L^*(s)$ gives the stated equality. The endpoint clamps alter it by only $O(\omega_n + \rho_n)$, and the normalization lower bounds $2/9 - O(\rho_n)$ and $1/3 - O(\rho_n)$ are uniform.

On $[2, 31/10]$ the map ρ_{KMS} is Lipschitz. The two explicit uniform denominator bounds above, Eq. (3), and bin variation therefore give an effective constant C_{KMS} for which every non-independent stratum has a genuine implemented vector coloring whose KMS exponent is at most

$$\sup_{s \in B^+} L^*(s) + \delta_{\text{KMS},n}, \quad \delta_{\text{KMS},n} := C_{\text{KMS}}\{\sqrt{\omega_n} + \rho_n\} = o(\varepsilon_n).$$

Enlarge C_0 and N_0 so that $\delta_{\text{KMS},n} \leq \varepsilon_n/4$. This proves the robust vector-coloring assertion before any use of KMS.

For a non-independent stratum, it remains to justify the stated success probability. Let

$$\ell_B := \sup_{s \in B^+} L^*(s), \quad m := |U|D^{-\ell_B - \varepsilon_n}.$$

If $U = \emptyset$, the claim is trivial. By Lemma 3.4 and $\delta_{\text{KMS},n} \leq \varepsilon_n/4$, one run returns a random independent set I with

$$\mathbb{E}[|I|] \geq c_{\text{KMS}}|U|D^{-\ell_B - \varepsilon_n/4}\{\log(n+2)\}^{-A_{\text{KMS}}}.$$

Because $\varepsilon_n \log D = \Theta(t^{5/3}) \gg \log \log n$, for all sufficiently large n ,

$$c_{\text{KMS}}\{\log(n+2)\}^{-A_{\text{KMS}}} \geq D^{-\varepsilon_n/4}.$$

Consequently

$$\mathbb{E}[|I|] \geq |U|D^{-\ell_B - \varepsilon_n/2} \geq 2m.$$

When $m \geq 1$, Lemma 4.3 gives

$$\Pr[|I| \geq m] \geq \frac{\mathbb{E}[|I|] - m}{|U| - m} \geq \frac{m}{|U|} = D^{-\ell_B - \varepsilon_n}.$$

Since $D = n^\beta$, $0 \leq \ell_B \leq 1/3$, and $\beta(1/3 + \varepsilon_n) < 1/2$ for all sufficiently large n , this probability is at least $n^{-1/2}$. If $0 < m < 1$, returning any vertex of U meets the claimed bound. Thus

$$R_B := \lceil (C+1)n^{1/2} \log n \rceil$$

fresh repetitions have failure probability at most $(1 - n^{-1/2})^{R_B} \leq n^{-(C+1)}$. This proves the lemma without assuming constant success probability. $\square$

We also use Ehrhard's inequality [Ehr83]. If $A, B \subseteq \mathbb{R}^m$ are convex, $0 \leq \theta \leq 1$, and γ is standard Gaussian measure, then

$$\Phi_{\text{cdf}}^{-1}(\gamma((1-\theta)A + \theta B)) \geq (1-\theta)\Phi_{\text{cdf}}^{-1}(\gamma(A)) + \theta\Phi_{\text{cdf}}^{-1}(\gamma(B)).$$

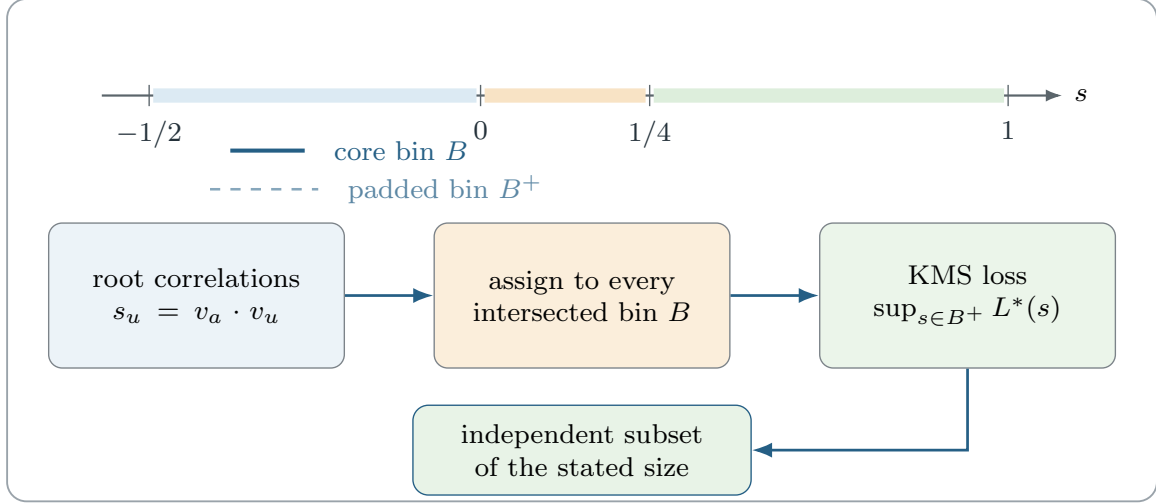


Figure 3: Uniform root-correlation rounding. Lemma 4.5 partitions the root correlations into padded bins separated by the three structural breakpoints and applies the appropriate KMS loss uniformly on each bin.

4.2 Matching menus and nested high-mass layers

Fix a selection parameter $q > 0$ and an induced graph H . Sample a standard Gaussian g , form $A_q := \{i : g \cdot v_i \geq \sqrt{qt}\}$, and choose a maximal matching of $H[A_q]$ by a fixed deterministic rule. Its unmatched vertices form an independent set. Define

$$d_{i,q}(H) := \Pr[i \text{ is unmatched} \mid i \in A_q], \quad \mu_{i,q}^H(j) := \Pr[ij \text{ is matched} \mid i \in A_q].$$

The matching probabilities supply the packing measure used to propagate neighborhood covers.

Lemma 4.6 (Matching packing). *The measure $\mu_{i,q}^H$ has mass $1 - d_{i,q}(H)$ and, for every $Y \subseteq N_H(i)$,*

$$\mu_{i,q}^H(Y) \leq \Pr[\max_{j \in Y} g \cdot e_{ij} \geq \sqrt{3qt}], \quad e_{ij} := \frac{2v_j + v_i}{\sqrt{3}} \perp v_i.$$

Proof. If ij is matched, both endpoint scores are at least $\sqrt{qt}$. Substitution in $v_j = -v_i/2 + \sqrt{3}e_{ij}/2$ gives the stated threshold. Conditioning on the score of v_i leaves its perpendicular Gaussian independent and standard. The total mass is the probability that i is matched, conditional on its selection. $\square$

We next account for the moment residuals in the matching-to-packing conversion.

Lemma 4.7 (Robust implemented matching packing). *The same statement holds for the unit directions formed from the rational moment data, with threshold $(\sqrt{3q} - O(\rho_n))t$, uniformly over the control menu. The continuous Gaussian experiment on these implemented directions defines the analytic packing; the rational random-bit experiment estimates its unmatched probability with the coupling error of Lemma 4.2.*

Proof. Put $c_{ij} := \hat{v}_i \cdot \hat{v}_j = -1/2 + O(\rho_n)$, $r_{ij} := \sqrt{1 - c_{ij}^2}$, and $\hat{e}_{ij} := (\hat{v}_j - c_{ij}\hat{v}_i)/r_{ij}$. This is an exact decomposition of the implemented directions. For large n , $c_{ij} < 0$, so selection of both endpoints

implies

$$g \cdot \hat{e}_{ij} \geq \frac{1 - c_{ij}}{r_{ij}} \sqrt{q} t = (\sqrt{3q} + O(\rho_n)) t.$$

The denominators and q lie in fixed compact ranges. Conditional orthogonal independence is exact for these directions. The transcript coupling changes the estimated probabilities only by the stated bias; it does not replace the continuous packing measure by a discrete measure. $\square$

To allow controls to depend on the root correlation, we prepare nested layers supporting the entire control menu.

Lemma 4.8 (Simultaneous nested high-mass layers). *Fix $J, C > 0$ and $\eta_0 > 0$. Let $\mathcal{M}_n$ be a deterministic menu of $n^{o(1)}$ pairs (q_k, H_k) in fixed positive compact ranges, with $q_k + H_k^2 \leq a_0 - \eta_0$. Procedure CONSTRUCTNESTEDLAYERS (Algorithm 2) either returns a verified independent set of size at least n^{1-a_0} , or returns $S_0 \supseteq \dots \supseteq S_J$, with $|S_J| = n - o(n)$, such that every $i \in S_r$ has every menu packing into S_{r-1} of mass at least $1 - \bar{\Phi}(H_k t)$. The alternative holds with probability at least $1 - n^{-C}$. Its ordinary-bit running time is polynomial.*

More generally, let $P \rightarrow \infty$ with $P \leq n^{O(1)}$, and put $T_P := \sqrt{2 \log P}$. Let a graph on $N \leq n$ vertices, with $N = P^{O(1)}$, have supplied unit vectors and a deterministic $P^{o(1)}$ menu with $q_k + H_k^2 \leq c$ in fixed positive compact ranges. If $N \geq P^{\tau+c+\epsilon}$ for fixed $\tau \geq 0$ and $\epsilon > 0$, the same marking procedure, with cutoff $NP^{-\epsilon/2}$, either finds a verified independent set of size at least $\lceil P^\tau \rceil$ or leaves $N - o(N)$ vertices after J levels. Every survivor has conditional matched mass at least $1 - \bar{\Phi}(H_k T_P)$ into the preceding level for every menu experiment. This mass conclusion needs no prescribed edge inner product; its conversion to a packing uses the supplied vectors.

Algorithm 2 Construct simultaneous nested high-mass layers

```

1: procedure CONSTRUCTNESTEDLAYERS( $G, J, C, \mathcal{M}_n, \eta_0$ ) ▷ Lemma 4.8
2:    $S_0 \leftarrow V(G)$ .
3:   for  $r = 1, \dots, J$  do
4:     for all  $k \in \mathcal{M}_n$  do
5:        $\delta_k \leftarrow \bar{\Phi}(H_k t)$ .
6:       Estimate every  $d_{i,q_k}(G[S_{r-1}])$  to error  $\delta_k/8$ .
7:        $B_k \leftarrow \{i \in S_{r-1} : \hat{d}_{i,k} \geq 3\delta_k/4\}$ .
8:       if  $|B_k| \geq n^{1-\eta_0/2}$  then
9:         Perform the polynomially many fresh matching trials in the proof.
10:        if a trial returns  $I$  with  $|I \cap B_k| \geq n^{1-a_0}$  then
11:          return the verified independent set  $I \cap B_k$ .
12:        else
13:          return FAILURE.
14:        end if
15:      end if
16:    end for
17:     $S_r \leftarrow S_{r-1} \setminus \bigcup_k B_k$ .
18:  end for
19:  return  $(S_0, \dots, S_J)$ .
20: end procedure

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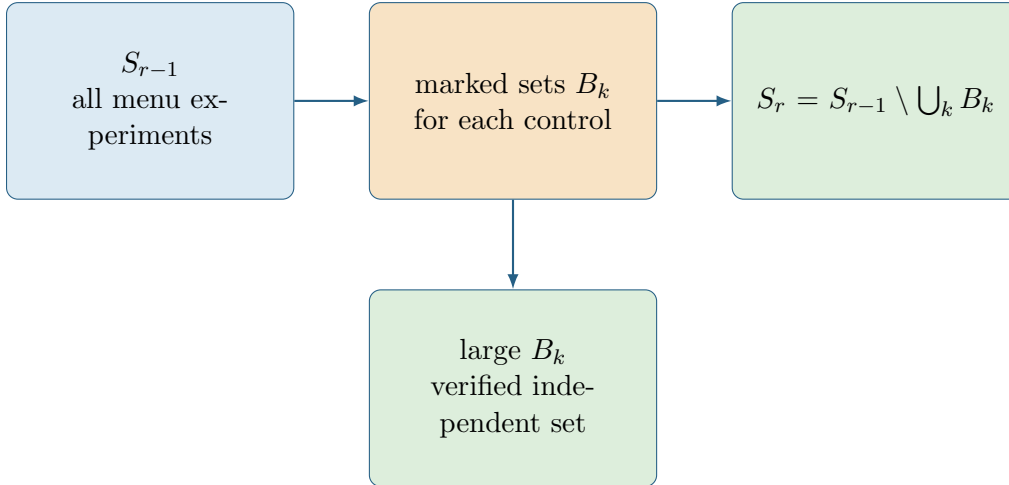
Proof. We prove the general assertion. Condition on the current induced graph and the complete past. Put $\delta_k := \bar{\Phi}(H_k T_P)$. On the simultaneous estimation event, a marked vertex has $d_{i,q_k} \geq 5\delta_k/8$, whereas an unmarked vertex has $d_{i,q_k} < 7\delta_k/8$. After the marked set B_k is fixed, a fresh matching trial returns an independent set I with

$$\mathbb{E}[|I \cap B_k| \mid \text{past}, B_k] \geq \frac{5}{8}|B_k|\bar{\Phi}(\sqrt{q_k}T_P)\bar{\Phi}(H_k T_P) \geq |B_k|P^{-c-o(1)}.$$

The first inequality sums the conditional unmatched probabilities; it uses no independence between different vertices. The second follows uniformly from Mills' inequalities. If $|B_k| \geq NP^{-\epsilon/2}$, the expectation is at least $P^{\tau+\epsilon/2-o(1)}$. Since $|I \cap B_k| \leq N \leq n$, Lemma 4.3 finds the target with polynomially many fresh trials and any fixed inverse-polynomial failure bound. Otherwise the menu removes at most $NP^{-\epsilon/2+o(1)} = o(N)$ vertices per level. Fixed J leaves $N - o(N)$ vertices with all required matched masses into the preceding layer, not necessarily into the new layer.

Use conditional rejection sampling and Hoeffding estimates. Both δ_k^{-1} and $\bar{\Phi}(\sqrt{q_k}T_P)^{-1}$ are uniformly polynomial. Choose the coupling bias below $\delta_k/100$. Each estimate and trial block receives failure budget $n^{-(C+4)}/(J|\mathcal{M}_n|)$. Conditioning on the past and then taking a union bound proves the probability and work claims. Certified fixed-factor brackets and vanishing threshold collars are absorbed by the fixed exponent reserves.

For the stated main construction take $P := n$, $N := n$, $\tau := 1 - a_0$, $c := a_0 - \eta_0$, and $\epsilon := \eta_0$. The cutoff is exactly that of Algorithm 2. Here the large-set expectation exceeds $n^{1-a_0+\eta_0/3}$ for large n , so $\lceil n^{a_0+1} \rceil$ fresh trials suffice. Finally, Lemmas 4.6 and 4.7 convert the matched masses into the asserted packings. $\square$



Every $i \in S_r$ has every menu packing into S_{r-1} .
If all marked sets are small, fixed-depth peeling leaves $n - o(n)$ vertices.

Figure 4: Simultaneous nested layers. Every vertex retained at level r has all menu packings into level $r - 1$. A large marked set yields verified independent-set progress.

4.3 Profile controls and tilted projection

The following controls allocate the Gaussian tail budget between selection, tilt, and projected failure.

Definition 4.9 (Profile controls). *Let U, z, h be the rational piecewise-affine profiles defined in Definition 5.4. Put*

$$e := 10^{-8}, \quad b := a_0 - e, \quad q(s) := b - z(s)^2 - h(s)^2,$$

$$\varrho(s) := \frac{\beta}{b}h(s), \quad H_p(s)^2 := z(s)^2 + h(s)^2 + e/2.$$

The control is $c(s) := (q(s), h(s), z(s), \varrho(s))$. For the first propagation use

$$z_1 := -\frac{7400728992}{10^{12}}, \quad h_1 := \frac{101675936082}{10^{12}}, \quad q_1 := b - z_1^2 - h_1^2, \quad \varrho_1 := \frac{\beta}{b}h_1.$$

Write $c_1 := (q_1, h_1, z_1, \varrho_1)$ and $H_{p,1}^2 := z_1^2 + h_1^2 + e/2$. The seed experiment has $q_{\text{seed}} := b$ and $H_{p,\text{seed}}^2 := e/2$.

Every propagation control has the exact reserves

$$q + H_p^2 = a_0 - e/2, \quad H_p^2 - z^2 - h^2 = e/2. \quad (9)$$

The seed has the same peeling cost. Lemma 5.5 proves that all controls lie in fixed positive compact ranges. Make the menu by evaluating the controls at a rational grid of mesh at most ω_n on $\mathcal{I}$ and at every knot, and include c_1 and the seed. Apply Lemma 4.8 with $\eta_0 := e/2$. At a parent choose the nearest grid point to its representative. Fixed Lipschitz bounds make this control differ from $c(s)$ by $O(\omega_n)$. The choice is made before considering any child.

For a root a and parent x set

$$s := v_a \cdot v_x, \quad r(s) := \sqrt{1 - s^2}, \quad w_x := \frac{v_x - sv_a}{r(s)}, \quad q_x := \frac{v_a - sv_x}{r(s)}.$$

An outgoing edge direction decomposes as $e_{xv} = yq_x + u_{xv}$, with $u_{xv} \perp v_a, w_x$ and $\|u_{xv}\|^2 = 1 - y^2$. For a control $c = (q, h, z, \varrho)$ define

$$b_c(y) := \sqrt{3q} - zy, \quad \eta := 9 \cdot 10^{-9}.$$

Conditioning near a chosen anchor score lets us project the packing while discarding fibers with negligible success probability.

Lemma 4.10 (Tilted projection and retention). *Let $T \rightarrow \infty$. Suppose a packing on unit directions $e_j = y_j q_x + u_j$, with $u_j \perp q_x$, has at most $\exp(\theta T^2/2)$ directions, mass at least $1 - \bar{\Phi}(H_p T)$, and threshold $(b_0 - o(1))T$. Put $b(y) := b_0 - zy$. Assume fixed positive lower bounds for $H_p^2 - z^2 - h^2$, h , η , and $b_0 - |z|$, and fixed upper bounds for all parameters. Retain the directions satisfying*

$$b(y)^2 \leq (\theta + h^2 + \eta)(1 - y^2).$$

Then, for a uniform $\epsilon_T = o(1)$,

$$\Pr[g \cdot u_j < (b(y_j) - \epsilon_T)T \text{ for every retained } j] \leq \bar{\Phi}(hT).$$

The threshold error is $O(T^{-1/2})$ plus the original packing error. Conservatively retaining additional directions preserves this conclusion.

Proof. Write the original Gaussian as $Zq_x + g_\perp$ and restrict Z to $|Z - zT| \leq \sqrt{T}$. This event has probability $\exp(-z^2T^2/2 + O(T^{3/2}))$. On it, complete projected failure implies failure of the original support after subtracting the stated threshold error. Independence therefore bounds complete projected failure by

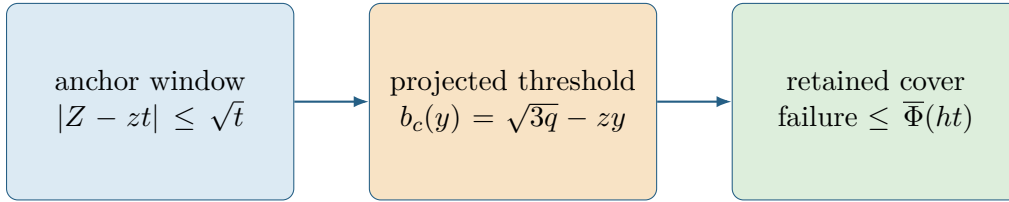
$$\exp(-(H_p^2 - z^2)T^2/2 + o(T^2)) = o(\bar{\Phi}(hT)).$$

The positive lower bound for $b(y)$ means that zero-norm directions never succeed. For a deleted positive-norm direction, its standardized threshold square, after the vanishing loss, exceeds $\theta + h^2 + \eta/2$ for large T , uniformly even for small norms. The union bound over all deleted directions is consequently $\exp(-(h^2 + \eta/2)T^2/2) = o(\bar{\Phi}(hT))$. Retained failure is contained in complete failure or success of a deleted direction. Their sum proves the assertion. $\square$

For the main process take $T := t$, $\theta := \beta$, $b_0 := \sqrt{3q}$, and $b := b_c$. Eq. (9) gives the required failure reserve, and Lemma 5.5 gives base positivity. Thus the main retention test remains

$$b_c(y)^2 \leq (\beta + h^2 + \eta)(1 - y^2), \quad (10)$$

with uniform threshold error $O(t^{-1/2} + \rho_n)$.



$$e_{xv} = yq_x + u_{xv}, u_{xv} \perp q_x; \text{ retain } b_c(y)^2 \leq (\beta + h^2 + 9 \cdot 10^{-9})(1 - y^2).$$

Figure 5: Tilted projection. Conditioning the anchor score near zt changes the projected threshold to $b_c(y)t$ and spends z^2 of the squared-tail budget. Only fibers satisfying the buffered retention test are needed.

4.4 Heterogeneous common-anchor composition

We use Ehrhard's inequality to strengthen the projected failure bound while retaining each fiber's own norm and threshold.

Lemma 4.11 (Quantitative common-anchor composition). *Let $T \rightarrow \infty$, let $N \leq \exp(\theta T^2/2)$, and put $L_j := \|u_j\| > 0$. Suppose the heterogeneous base thresholds b_j satisfy*

$$\Pr[g \cdot u_j < (b_j - \epsilon_T)T \text{ for every } j] \leq \bar{\Phi}(hT), \quad \epsilon_T = o(1).$$

Let $v, h, \varrho > 0$ and all parameters lie in fixed compact ranges, with v, h, ϱ bounded below. For bounded $\xi \geq 0$ put

$$d_j := \frac{v}{\varrho + h}(L_j \sqrt{\theta + \varrho^2} - b_j), \quad Q_j(\xi) := vb_j - d_j(\xi - h)_+.$$

There are uniform errors $\epsilon'_T = o(1)$ and $\epsilon''_T = o(1)$ such that

$$\Pr[v g \cdot u_j < (Q_j(\xi) - \epsilon'_T)T \text{ for every } j] \leq \bar{\Phi}((\xi - \epsilon''_T)T).$$

Proof. For $\xi \leq h$ use the base-failure hypothesis. Otherwise set $\alpha := (\xi - h)/(\varrho + h)$ and $\vartheta := (\xi - h)/(\xi + \varrho)$. Replace b_j temporarily by $b'_j := b_j - \epsilon_T$ and put $c_j := L_j \sqrt{\theta + \varrho^2}$. Consider the convex simultaneous-failure sets with thresholds $b'_j - \alpha(c_j - b'_j)$, b'_j , and c_j , denoted by $\mathcal{C}_0, \mathcal{C}_b, \mathcal{C}_\varrho$. The coordinatewise identity

$$(1 - \vartheta)(b'_j - \alpha(c_j - b'_j)) + \vartheta c_j = b'_j$$

gives $(1 - \vartheta)\mathcal{C}_0 + \vartheta\mathcal{C}_\varrho \subseteq \mathcal{C}_b$. The high-anchor union bound and Mills' inequalities give

$$\Phi_{\text{cdf}}^{-1}(\Pr[\mathcal{C}_\varrho]) \geq \varrho T - O(\log T/T), \quad \Phi_{\text{cdf}}^{-1}(\Pr[\mathcal{C}_b]) \leq -hT.$$

Ehrhard's inequality applied to the inclusion yields

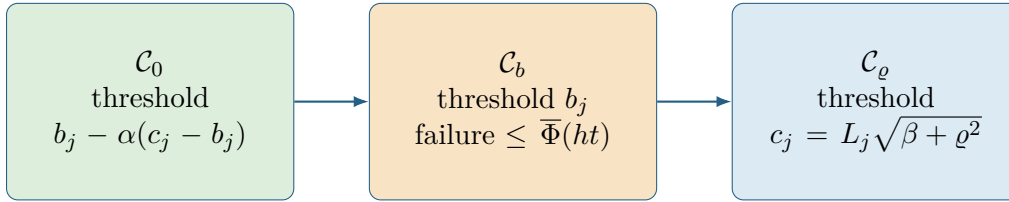
$$\Phi_{\text{cdf}}^{-1}(\Pr[\mathcal{C}_0]) \leq -\xi T + O(\log T/T).$$

Here $(h + \vartheta\varrho)/(1 - \vartheta) = \xi$, and $\vartheta/(1 - \vartheta) = \alpha$ is uniformly bounded. Restoring b_j costs $(1 + \alpha)\epsilon_T = o(1)$ in threshold. Multiplication by v proves the claim. The same interpolation coefficient is used for every fiber; neither equal norms nor equal base thresholds were assumed. $\square$

For the main process, Lemma 4.10 supplies the base hypothesis. Apply Lemma 4.11 with $T := t$, $\theta := \beta$, $v := \sqrt{3}/2$, and $b_j := b_c(y_j)$. Thus, for $0 \leq \xi \leq \sqrt{1 + e}$, define

$$d_c(y) := \frac{\sqrt{3}}{2(\varrho + h)}(\sqrt{1 - y^2}\sqrt{\beta + \varrho^2} - b_c(y)), \quad Q_c(y, \xi) := \frac{\sqrt{3}}{2}b_c(y) - d_c(y)(\xi - h)_+.$$

Lemma 5.5 gives $\varrho^2 > h^2 + e$, so Eq. (10) implies $d_c(y) > 0$.



$$(1 - \vartheta)\mathcal{C}_0 + \vartheta\mathcal{C}_\varrho \subseteq \mathcal{C}_b$$

One common coefficient; heterogeneous fiber thresholds.

Figure 6: Common-anchor composition. The same interpolation coefficient is used in every coordinate of three convex failure sets, despite their heterogeneous thresholds.

We now combine a parent cover with its outgoing fibers; a cardinality reserve controls the total probability of failed propagation.

Lemma 4.12 (Uniform propagation of a covered state). *Let $T \rightarrow \infty$. Suppose unit vectors have edge inner product $-k$, where $0 < k < 1$, and put $v := \sqrt{1 - k^2}$ and $r(s) := \sqrt{1 - s^2}$. Fix a root a and a deterministic parent family X . For an edge xz put*

$$s_x := v_a \cdot v_x, \quad \tau_z := v_a \cdot v_z, \quad y_{xz} := \frac{\tau_z + k s_x}{v r(s_x)}, \quad A_{xz} := \frac{k + s_x \tau_z}{r(s_x)}.$$

Assume A_{xz} and all normalization denominators have fixed positive lower bounds, and all coefficients are uniformly bounded. Choose the control before identifying a child. Suppose the retained

perpendicular fibers at each x satisfy Lemma 4.11, with contributions $Q_{xz}(\xi)$ and uniform vanishing errors. Let $\lambda \geq 0$ be bounded, and suppose

$$\Pr[\max_{x \in X} g \cdot w_x \geq (\lambda - o(1))T] \geq \exp(-o(T^2)), \quad w_x := \frac{v_x - s_x v_a}{r(s_x)}.$$

For bounded caps $m_x \geq \lambda^2$, assume a fixed $\epsilon > 0$ such that $|X| \leq \exp((m_x - \epsilon)T^2/2)$ for every $x \in X$. Partition all retained pairs into $\exp(o(T^2))$ deterministic child families. At least one family is covered with probability $\exp(-o(T^2))$ at the minimum of its pair thresholds

$$\frac{A_{xz}\lambda + Q_{xz}(\sqrt{m_x - \lambda^2})}{r(\tau_z)} - o(1).$$

A negative minimum may instead be replaced by zero.

Proof. Put $q_x := (v_a - s_x v_x)/r(s_x)$ and write the normalized edge direction as $e_{xz} = y_{xz}q_x + u_{xz}$, where $u_{xz} \perp v_a, w_x$. Substitution in $v_z = -kv_x + ve_{xz}$ gives

$$\tau_z = -ks_x + vr(s_x)y_{xz}, \quad v_z - \tau_z v_a = -A_{xz}w_x + vu_{xz}. \quad (11)$$

Draw a Gaussian in $v_a^\perp$. For a fixed x , the reflected score $g \cdot (-w_x)$ and the perpendicular fiber Gaussian are independent. Put $\xi_x := \sqrt{m_x - \lambda^2}$. The common-anchor bound makes the probability of outer success with failure of every retained fiber at most

$$\bar{\Phi}((\lambda - o(1))T) \bar{\Phi}((\xi_x - o(1))T).$$

For bounded nonnegative b , $(b - o(1))_+^2 \geq b^2 - o(1)$. Since $\lambda^2 + \xi_x^2 = m_x$, summing over X and using the cardinality reserve bounds all bad outer successes by $\exp(-\epsilon T^2/2 + o(T^2))$. No independence across parents is used. This error is negligible relative to the reflected parent cover. Eq. (11), positivity of A_{xz} , and the lower radius bound give the asserted child thresholds. Deterministic binning loses only $\exp(o(T^2))$ in cover probability. Taking a lower minimum preserves the cover. At threshold zero a nonempty family of unit vectors is covered with probability at least $1/2$.

All these estimates are uniform under vanishing perturbations of the edge identities, coefficients, and bin representatives, provided the stated denominator bounds and fixed reserves remain valid. Lemma 4.2 supplies precisely these perturbations in the applications below. $\square$

4.5 A hierarchy of local rounding procedures

The root-correlation rounding procedure can itself be used inside a local matching chain. We first construct a zero-tilt chain whose intermediate strata are rounded by quantitative KMS. We then use that procedure to round the intermediate strata of a second chain, with a tilt chosen from the parent correlation. A third chain uses both preceding procedures. Two further tilted chains use the first two procedures on their intermediate strata and a smaller upper bound on the specified target exponent. One of them also tilts the first propagation, without changing seed creation. Every statement below concerns a specified independent-set target, not an unrestricted independent-set ratio guarantee.

We encode the vertex costs of these procedures in the following loss functions.

Definition 4.13 (Local costs and fallback losses). Put $u := 10^{-6}$, $\delta := 10^{-3}$, and define

$$x_0(L) := \frac{61}{100}L + \frac{13}{25}L^2, \quad x_1(L) := \frac{19}{2000} + \frac{221}{500}L + \frac{427}{500}L^2.$$

$$x_2(L) := \frac{17}{2000} + \frac{221}{500}L + \frac{427}{500}L^2.$$

Define also

$$x_4(L) := \frac{1}{125} + \frac{221}{500}L + \frac{427}{500}L^2, \quad x_5(L) := \frac{14}{25}L + \frac{1}{2}L^2, \quad T_* := \frac{1247}{1000}.$$

For $0 \leq L \leq 1/3$, put

$$\begin{aligned} P_{\text{ext}}(L) &:= \min\{L, x_0(\max\{L, 1/50\})\}, & P_0(L) &:= \min\{L, x_0(\max\{L, 1/10\}) + \delta\}, \\ P(L) &:= \min\{x_0(\max\{L, 1/50\}), x_1(\max\{L, 1/10\})\}, & \sigma_{\text{sub}} &:= 10^{-6}, \\ G(L) &:= \min\{P_{\text{ext}}(L), x_1(\max\{L, 3/100\}), x_2(\max\{L, 4/25\}), \\ &\quad x_4(\max\{L, 23/100\}), \hat{x}_5(L)\}, \end{aligned}$$

where

$$\hat{x}_5(L) := x_5(\max\{L, 13/200\}) + (L - 7/50)_+.$$

These formulas also define the losses on a fixed small collar above $1/3$. The internal cap of chain 1 retains the charge δ in P_0 . The final loss G uses the five local procedures with an arbitrarily small fixed positive sufficient-cardinality surplus, as proved below; it does not remove δ from the internal scalar caps or σ_{sub} from the subordinate charges in Q_2, Q_4, Q_5 .

We use an ambient order n , a degree bound $D = n^\theta$, and $S := \sqrt{2 \log D}$, where θ lies in a fixed positive compact interval. A supplied approximate strict loss- L vector coloring consists of unit vectors with

$$v_i \cdot v_j = c(L) + O(\eta_n), \quad c(L) := -\frac{1-L}{1+L} \quad (ij \in E), \quad (12)$$

uniformly, where $\eta_n = o(1)$ has an effective common bound. The vectors have polynomial-size Gram data and certified computable enclosures, and their Gaussian and vector operations use the ordinary-bit implementation of Lemma 4.2. In our application they are explicit normalized combinations of the rational moment vectors. Thus this input condition does not posit exact-real access to arbitrary vectors. Every error statement below is uniform after the fixed depths, work bound, and failure exponent have been chosen.

Projection improves the loss inside a correlation bin, while weakening places that loss in a subordinate procedure's supported range.

Lemma 4.14 (Strict projection and weakening). *Suppose a signed root-correlation bin has diameter tending to zero and its root-perpendicular normalization is bounded away from zero. Projecting the vectors in Eq. (12) perpendicular to the root gives an approximate strict coloring with loss*

$$L_{\text{pr}}(L, s) := \max\{0, L - (1+L)s^2\},$$

up to a uniform vanishing outward collar. If the unclamped expression is separated below zero, the bin is independent. Near zero one obtains a genuine vector $2 + o(1)$ coloring. Moreover, for $0 \leq L \leq W$, a strict loss- L coloring can be weakened to strict loss W without new moment data.

Proof. For exact equal correlations s with a unit root v_a , put $w_i := (v_i - sv_a)/\sqrt{1-s^2}$. On an edge,

$$w_i \cdot w_j = \frac{c(L) - s^2}{1 - s^2}, \quad \frac{1 + w_i \cdot w_j}{1 - w_i \cdot w_j} = L - (1+L)s^2.$$

A negative right-hand side would force an edge inner product below -1 . For a narrow bin, first use the exact implemented correlations in the projection; the positive normalization bound then gives the asserted uniform error. Clamp a loss collar at zero and apply Lemma 3.4 with a genuine parameter at least 2. For weakening, let e be a common unit vector perpendicular to all v_i and set

$$\tilde{v}_i := \sqrt{\frac{1+L}{1+W}}v_i + \sqrt{\frac{W-L}{1+W}}e.$$

These vectors are unit and have edge inner product $(W-1)/(W+1)$. The same calculation preserves uniform vanishing edge errors. $\square$

We specify the controls and cardinality caps that determine each local chain.

Definition 4.15 (Local controls and transition). *The chain indices are $\mathcal{K} := \{0, 1, 2, 4, 5\}$, with loss ranges*

$$\begin{aligned}\mathcal{L}_0 &:= [1/50, 17/50], & \mathcal{L}_1 &:= [3/100, 167/500], & \mathcal{L}_2 &:= [4/25, 1/3], \\ \mathcal{L}_4 &:= [23/100, 1/3], & \mathcal{L}_5 &:= [13/200, 7/50].\end{aligned}$$

Put $J_0 := 16$, $J_1 := 32$, and $J_2 := J_4 := J_5 := 128$. For $i \in \mathcal{K}$, put $x := x_i(L)$ and define

$$k := \frac{1-L}{1+L}, \quad v := \frac{2\sqrt{L}}{1+L}, \quad q := \frac{x(1+x)}{1+L}, \quad H^2 := \frac{x(L-x)}{1+L}, \quad b_0 := \sqrt{q/L}, \quad r(s) := \sqrt{1-s^2}.$$

Chain 0 uses $z(s) := 0$, $h(s) := H$, and $\varrho(s) := H/x$. For chains 1, 2, 4, and 5 define

$$\chi(L) := \min\{1, \max\{1/2, 5/2 - 6L\}\}, \quad z(s) := \text{sgn}(s) \min\{17H/20, \chi(L)L\sqrt{q}|s|/r(s)\},$$

where $z(0) := 0$, and put

$$h(s) := \sqrt{H^2 - z(s)^2}, \quad \varrho(s) := h(s)/x.$$

The first propagation of chains 0, 1, 2, 4 instead uses $z := 0$, $h := H$, and $\varrho := H/x$. Chain 5 uses its parent-dependent control on the first propagation as well. For a parent correlation s and a child correlation t , define

$$\begin{aligned}y &:= \frac{t+ks}{vr(s)}, & A &:= \frac{k+st}{r(s)}, & b_s(y) &:= b_0 - z(s)y, \\ d_{L,s}(y) &:= \frac{v}{\varrho(s) + h(s)} (\sqrt{1-y^2} \sqrt{1+\varrho(s)^2} - b_s(y)).\end{aligned}$$

For $m \geq \lambda^2$ put

$$F_{i,m}(L, s, \lambda, t) := \frac{A\lambda + vb_s(y) - d_{L,s}(y)(\sqrt{m-\lambda^2} - h(s))_+}{r(t)}. \quad (13)$$

The initial cap is $m_1 := 1 + u$. Put $T_i := 5/4$ for $i \in \{0, 1, 2\}$ and $T_4 := T_5 := T_*$. Later caps are

$$m_i(L, s) := T_i + \min\{x_i(L) + \delta, Q_i(L_{\text{pr}}(L, s))\} + u, \quad (14)$$

Here $Q_0(w) := w$, $Q_1(w) := P_0(w)$, and $Q_2(w) := Q_4(w) := Q_5(w) := \min\{w, P(w) + \sigma_{\text{sub}}\}$. For the subordinate calls, define $\mathcal{C}_0(\ell) := \emptyset$,

$$\mathcal{C}_1(\ell) := \{(0, \max\{\ell, 1/10\}, \delta)\},$$

$$\mathcal{C}_2(\ell) := \{(0, \max\{\ell, 1/50\}, \sigma_{\text{sub}}/2), (1, \max\{\ell, 1/10\}, \sigma_{\text{sub}}/2)\},$$

and $\mathcal{C}_4(\ell) := \mathcal{C}_5(\ell) := \mathcal{C}_2(\ell)$. Each triple specifies the subordinate index, weakened loss, and fixed surplus.

All signed bins in this definition are bins for the actual graph procedure. Only the scalar certificate may use absolute correlations as a relaxation. Lemma 5.2 supplies positive normalization, base-threshold, and composition margins on the reachable domain. In particular $A, d_{L,s} > 0$, so the transition is increasing in its nonnegative parent threshold and decreasing in its cap.

The finite chain certificates turn these recurrences into rounding procedures for a prescribed independent-set target.

Theorem 4.16 (Specified-target local rounding). *Let $\mathcal{K}, \mathcal{L}_i, T_i$ be as in Definition 4.15. Fix $C > 0$, $\epsilon_0 > 0$, $i \in \mathcal{K}$, and $1 \leq t_0 \leq T_i$. The following holds uniformly for $L \in \mathcal{L}_i$ and all sufficiently large D . If a graph of maximum degree at most D has supplied data as in Eq. (12) and at least*

$$\lceil D^{t_0+x_i(L)+\epsilon_0} \rceil$$

vertices, the corresponding randomized ordinary-bit polynomial-time procedure finds an independent set of size at least $\lfloor D^{t_0} \rfloor$ with probability at least $1 - n^{-C}$. The polynomial work bound and the sufficiently-large-input threshold may depend on the fixed ϵ_0 . The finite termination assertions used here are proved in Lemma 5.3.

Proof. We first prove chain 0, using only quantitative KMS for intermediate rounding. We then prove chain 1 using chain 0, and finally chains 2, 4, and 5 using only chains 0 and 1. The last three do not call one another. This order makes the dependence finite. The same argument below applies to all five chains.

Put

$$\alpha := \min\{\epsilon_0/2, \delta/4\}, \quad \nu := \alpha/4, \quad \epsilon_{\text{peel}} := \alpha - \nu > 0.$$

First restrict to an induced graph of order N with

$$D^{t_0+x+\alpha} \leq N \leq 3D^{t_0+x+\alpha}. \quad (15)$$

For example, compute a certified upper approximation B to $D^{t_0+x+\alpha}$ within a factor two and keep $\min\{|V|, \lceil B \rceil\}$ vertices. The size promise gives the lower bound and the upper bound holds for large D , since $\epsilon_0 - \alpha > 0$. This step preserves the degree and vector assumptions and avoids deciding an exact-real ceiling. The upper bound in Eq. (15), not merely the original size promise, will be used for counting closure.

The outgoing experiment has selection exponent q and unmatched-tail parameter $\sqrt{H^2 + \nu}$. The separate seed experiment has selection exponent x and unmatched-tail parameter $\sqrt{\nu}$. Both cost $x + \nu$, because $q + H^2 = x$. Apply the general form of Lemma 4.8 with

$$P := D, \quad \tau := t_0, \quad c := x + \nu, \quad \epsilon := \alpha - \nu.$$

Eq. (15) verifies its size hypothesis. It also gives $N = D^{O(1)}$ and $D \leq n$, uniformly over the fixed parameter ranges. The cutoff is $ND^{-(\alpha-\nu)/2}$; certified fixed-factor brackets are absorbed by the strict exponent gap. Thus either a fresh trial finds the target or each fixed level loses only $o(N)$ vertices. Use the layer budget J_i from Definition 4.15 to construct the outgoing layers $S_0 \supseteq \dots \supseteq S_{J_i}$ and run the seed marking step inside S_{J_i} . For this last step put $N' := |S_{J_i}| = N - o(N)$ and use $\epsilon := (\alpha - \nu)/2$ in the general marking lemma. For sufficiently large D ,

$$N' \geq D^{t_0+x+\nu+(\alpha-\nu)/2},$$

because Eq. (15) has the strictly larger exponent $t_0 + x + \alpha$. Thus the seed step has its own valid size reserve and cutoff $N'D^{-(\alpha-\nu)/4}$. In the absence of progress there is an unmarked root a whose

seed neighborhood lies in S_{J_i} ; every such neighbor has its outgoing packing into S_{J_i-1} . At each subsequent step the available layer index decreases by one. No packing into a newly formed layer is asserted.

For an edge with exact inner product $-k$, simultaneous selection of its endpoints above $\sqrt{q}S$ forces perpendicular score at least $(1+k)\sqrt{q}S/v = b_0S$. The seed experiment consequently gives a cover at threshold $(\sqrt{x/L} - o(1))S$, with at most D distinct neighbors and root correlation $-k + o(1)$. The approximate case follows from the exact implemented-edge decomposition and the uniform denominator bounds. The seed failure exponent $\nu > 0$ makes its cover probability at least $1/2$ for large D ; its counting cap remains $1 + u$.

For chain 5, distinguish creation of this seed cover from its first propagation. Every seed endpoint already has an outgoing packing in the preceding layer. We choose the tilt from the parent correlation before selecting a successful child, and condition that outgoing experiment near $z(s)S$. Its failure exponent becomes

$$H^2 + \nu - z(s)^2 = h(s)^2 + \nu > h(s)^2.$$

The equality uses $H^2 = z(s)^2 + h(s)^2$, and the inequality uses $\nu > 0$. The cover being propagated still has at most D endpoints. Therefore the tilted-projection and common-anchor arguments below apply with first cap $1 + u$, including both child orientations. They impose no condition that the propagation control equal the control that created the seed cover.

Choose signed bins of width $\omega_D := 2^{-\lceil S \rceil}$ and an error scale $\epsilon_D := S^{-1/3} + \sqrt{\eta_n} = o(1)$, enlarging the effective η_n bound to include the smaller numerical errors if necessary. The number of bins is $D^{o(1)}$, and $\log(1/\omega_D) = o(\epsilon_D S^2)$. A covered state consists of a deterministic nonempty endpoint set X in one such bin, its representative correlation s , and a nonnegative threshold λ , with

$$\Pr[\max_{x \in X} g \cdot w_x \geq (\lambda - \epsilon_D)S] \geq \exp(-c_j \epsilon_D S^2)$$

at fixed depth j . The algorithm never tests this probability. It keeps every deterministic candidate produced by the bin enumeration.

We next justify the later-state caps. In every chain, apply quantitative KMS to the projected vectors of every candidate bin. In chain 1, also apply chain 0 with fixed surplus δ to those vectors after weakening their loss to at least $1/10$. Make this call only if its sufficient-cardinality threshold is met; otherwise the corresponding cardinality bound already holds. Its parameter stays below $17/50$, since a projected loss is at most $L + o(1) \leq 167/500 + o(1)$. In chains 2, 4, and 5, also apply chains 0 and 1 with fixed surplus $\sigma_{\text{sub}}/2$, after weakening their losses to at least $1/50$ and $1/10$, respectively. The remaining $\sigma_{\text{sub}}/2$ absorbs their uniform vanishing loss collars and rounding errors. These calls have parameters at most $1/3 + o(1)$, hence lie below $17/50$ and $167/500$, respectively, for large D . The lowered zero-tilt range is certified in Lemma 5.3. Only calls satisfying their sufficient-cardinality promises are made. Every subordinate call uses the same target $\lfloor D^{t_0} \rfloor$, and builds its own layers inside that induced subgraph. Lemma 4.14 and the uniform KMS bound imply that, on the simultaneous good event, failure to return the target gives, with a fixed $u/4$ cushion for outward loss collars and rounding errors,

$$|X| \leq D^{t_0 + Q_i(L, s) + u/4 + o(1)}.$$

For chain 1 the δ cost of the subordinate call is already included in P_0 ; the lowered parent range does not lower that subordinate range. Combining this bound with Eq. (15), $\alpha \leq \delta/4$, and $t_0 \leq T_i$ gives, uniformly on the padded bin,

$$|X| \leq D^{m_i(L, s) - u/2} \tag{16}$$

for sufficiently large D . Indeed the two exponents are bounded by $t_0 + x_i(L) + \alpha + o(1)$ and $t_0 + Q_i(L_{\text{pr}}(L, s)) + u/4 + o(1)$, respectively, whereas the internal cap uses $x_i(L) + \delta$ and has the additional reserve u . Neither this δ nor the one in P_0 has been removed; the subordinate charge σ_{sub} in Q_2, Q_4, Q_5 is also retained. This is a consequence on the simultaneous good event, not a deterministic inference from randomized failure.

We now apply the shared Gaussian lemmas in degree units. In Lemma 4.10 take

$$T := S, \quad \theta := 1, \quad b_0 := \sqrt{q/L}, \quad H_p^2 := H^2 + \nu, \quad \eta := u/2.$$

The control is chosen from the parent representative before any successful child is identified. The projection reserve is $H_p^2 - z(s)^2 - h(s)^2 = \nu > 0$; it need not be as large as u . The actual procedure conservatively retains pairs meeting

$$b_s(y)^2 \leq (1 + h(s)^2 + u/2)(1 - y^2). \quad (17)$$

The fixed gap from $u/2$ to the certificate allowance u includes every boundary-straddling retained pair for large D . Lemma 5.2 supplies base positivity and all normalization and composition bounds. Lemma 4.11 with $v := 2\sqrt{L}/(1 + L)$ gives

$$Q_{xz}(\xi) := vb_s(y) - d_{L,s}(y)(\xi - h(s))_+.$$

Apply Lemma 4.12 with $k := (1 - L)/(1 + L)$ and $m_x := m_i(L, s_x)$. Eq. (16) gives $\epsilon := u/2$. For the first propagation instead use $|X| \leq D$, $m_x := 1 + u$, and $\epsilon := u$. Its pair threshold is exactly Eq. (13) up to a uniform $o(1)$. The signed bins satisfy $\log(1/\omega_D) = o(\epsilon_D S^2)$, so the proof preserves the displayed covered-state invariant at the next depth after increasing its fixed constant c_j .

Enumerate every retained graph pair, place its child in every signed bin met by its certified correlation interval, and count distinct vertex IDs. Store a conservative minimum of the pair thresholds, with a deterministic minimum-attaining witness, as in Algorithm 4. The shared lemma justifies both the covered child and the nonnegative clamp. Run all intermediate rounding calls before testing raw counting closure. On the good event, Eq. (16) and the raw cover prevent a covered candidate from being discarded. Downward clipping to a certified lower cap costs only $o(1)$, by the same three-term argument as Lemma 4.22. Control approximation, witness-to-bin representatives, implemented geometry, threshold evaluation and clipping jointly cost less than 10^{-5} per step for sufficiently large D . The initial correlation error is below 10^{-8} and its threshold loss below 10^{-5} . Square roots have a uniform square-root modulus at a zero radicand; no bounded derivative there is assumed.

The resulting continuing covered branch is among the trajectories excluded by Lemma 5.3. It cannot reach depths 10, 28, 37, 40, and 25 in chains 0, 1, 2, 4, and 5, respectively, before the available layers are exhausted. Hence some call returns the verified target independent set. At these fixed depths the bin and branch counts are $n^{o(1)}$, and each matching, enumeration and rounding call has polynomial work. Subordinate calls follow the order specified at the start of the proof; no chain calls itself or the main coloring theorem. Fix ϵ_0 , and hence α and ν , before choosing the total work exponent, then the ordinary-bit precision and conditional failure budgets. A union bound over the complete adaptive execution gives failure at most n^{-C} . $\square$

Write AMPLIFIEDKMS for independent repetitions of Algorithm 1, retaining the largest verified independent set. Its polynomial repetition budget is fixed by Lemma 4.3 and the simultaneous failure budget for the complete execution.

We combine the five procedures by taking the smallest vertex cost for their common target.

Algorithm 3 Specified-target local chain

```
1: procedure LOCALCHAIN( $i, H, D, L, t_0, \epsilon_0, C$ ) ▷ Theorem 4.16
2:   Set  $x \leftarrow x_i(L)$ ,  $T \leftarrow \lfloor D^{t_0} \rfloor$ , and  $J \leftarrow J_i$  from Definition 4.15.
3:   Set  $\alpha \leftarrow \min\{\epsilon_0/2, \delta/4\}$  and  $\nu \leftarrow \alpha/4$ .
4:   if a certified size comparison excludes the sufficient-cardinality promise then
5:     return FAILURE.
6:   end if
7:   Restrict  $H$  to the prescribed order in Eq. (15).
8:    $\mathcal{R} \leftarrow$  the general marking procedure of Lemma 4.8, constructing  $J$  outgoing layers with
   unmatched-tail exponent  $x - q + \nu$ .
9:   if  $\mathcal{R}$  is a verified independent set of size at least  $T$  then
10:    return  $\mathcal{R}$ .
11:  else if  $\mathcal{R} = \text{FAILURE}$  then
12:    return FAILURE.
13:  end if
14:   $(S_0, \dots, S_J) \leftarrow \mathcal{R}$ .
15:  Run the seed marking step with unmatched-tail exponent  $\nu$ ; return any verified target set.
16:  if the seed step returns FAILURE or no unmarked root remains then
17:    return FAILURE.
18:  end if
19:  Choose an unmarked root and initialize all of its signed seed-bin states.
20:  for  $j = 1, \dots, J - 1$  do
21:    for all current endpoint states do
22:      Choose the first or parent-selected control from Definition 4.15.
23:      Enumerate all retained pairs into  $S_{J-j}$  and form every signed raw child bin.
24:      for all raw child bins do
25:        Let  $Y$  be the bin; form projected vectors  $(w_y)$  with outward loss  $\ell$ .
26:         $I \leftarrow \text{AMPLIFIEDKMS}(H[Y], (w_y)_{y \in Y}, 2/(1 - \ell), D)$ . ▷ Algorithm 1
27:        for all  $(k, W, \epsilon) \in \mathcal{C}_i(\ell)$  do
28:          if the chain- $k$  size threshold with surplus  $\epsilon$  is met then
29:            Form  $H'$  with the weakened loss- $W$  vectors.
30:             $I_k \leftarrow \text{LOCALCHAIN}(k, H', D, W, t_0, \epsilon, C')$ . ▷ Algorithm 3
31:            Keep the larger verified independent set among  $I$  and  $I_k$  in  $I$ .
32:          end if
33:        end for
34:        if  $|I| \geq T$  then
35:          return  $I$ .
36:        end if
37:        Test raw counting closure; discard only candidates failing this test.
38:        Clip each remaining threshold downward to its bin cap and retain the state.
39:      end for
40:    end for
41:  end for
42:  return FAILURE.
43: end procedure
```

Lemma 4.17 (Loss supplied to the main state cap). *Fix $C > 0$, $\epsilon_0 > 0$, and $1 \leq t_0 \leq T_*$. On an approximate strict loss- L bin, $0 \leq L \leq 1/3$, $\lceil D^{t_0+G(L)+\epsilon_0} \rceil$ vertices suffice, for sufficiently large D , for the combination of quantitative KMS and the five local procedures to return the specified target. On the simultaneous good event, failure to return $\lfloor D^{t_0} \rfloor$ therefore implies*

$$|X| \leq D^{t_0+G(L)+\epsilon_0}.$$

For an integer main target $\lceil n^\alpha \rceil$, with $1 < \alpha/\beta < T_$ and $D = n^\beta$, choose any fixed sufficiently small $\sigma > 0$ and set $t_0 := (\alpha + \sigma)/\beta$. Then the local output meets that target for large n , while the sufficient-cardinality exponent is $\alpha + \beta G(L) + \sigma + \beta \epsilon_0$. The fixed positive σ and ϵ_0 may be chosen inside the reserve of the main cap. The work bound and input threshold may depend on these fixed choices; no claim uniform as ϵ_0 tends to zero is needed.*

Proof. For chains 0, 1, and 2, weaken losses to at least $1/50$, $3/100$, and $4/25$, respectively. The chain-1 range in Theorem 4.16 supports this weakening floor; its internal subordinate loss P_0 and the loss P used by chain 2 remain unchanged. For chains 4 and 5, weaken losses to at least $23/100$ and $13/200$, respectively. Chain 4 is supported through $1/3$ with a fixed collar. Use chain 5 only through $7/50$. This does not change G , since

$$x_5(7/50) - x_1(7/50) = \frac{51}{625000} > 0.$$

The equality follows by substitution, and the inequality is immediate. Above $7/50$, the derivative of $\hat{x}_5$ is at least $17/10$, whereas $x'_1(L) \leq 1517/1500$ on $[0, 1/3]$. Thus x_1 dominates the unsupported part of $\hat{x}_5$, also on a sufficiently small collar. Lemma 4.14 and Theorem 4.16 give the five supported polynomial costs in Definition 4.13, and KMS gives cost L . Running all size-qualified calls with local surplus $\epsilon_0/2$ gives their minimum; the remaining $\epsilon_0/2$ absorbs their uniform vanishing loss collars, integer and logarithmic factors. They use the same target, so their cardinality implications can be intersected; no ratio bound is inferred. Finally, $\lfloor D^{t_0} \rfloor = \lfloor n^{\alpha+\sigma} \rfloor \geq \lceil n^\alpha \rceil$ for large n . $\square$

We apply the combined local loss to root strata at the target required by the main algorithm.

Lemma 4.18 (Enhanced root-stratum rounding). *Let G be the local loss from Definition 4.13, and let X be a padded root-correlation stratum as in Lemma 4.5. Put $e := 10^{-8}$ and*

$$t_0 := \frac{1 - a_0 + e/8}{\beta}, \quad \epsilon_{\text{loc}} := e/16.$$

For every fixed $C > 0$, procedure ROUNDROOTSTRATUM returns a verified independent set of size at least $\lceil n^{1-a_0} \rceil$ with probability at least $1 - n^{-C}$ whenever

$$|X| \geq n^{1-a_0+\beta \sup_{s \in B^+} G(L^*(s))+e/4}. \quad (18)$$

It runs quantitative KMS and the five local procedures on the corresponding strict vector-coloring stratum, retaining the largest verified output. The running time is polynomial, uniformly over the bins.

Proof. The actual exponent gives

$$1 < \frac{1 - a_0}{\beta} < t_0 = \frac{3339090009}{2678180008} < T_*,$$

where the equality substitutes a_0, β, e , and the last inequality follows from $T_*\beta - (1 - a_0 + e/8) = 37528811/250000000000 > 0$. The first two inequalities use $a_0, e > 0$. For large n , $\lfloor D^{t_0} \rfloor =$

$\lfloor n^{1-a_0+e/8} \rfloor \geq \lceil n^{1-a_0} \rceil$. Lemma 4.5 supplies approximate strict loss- $L^*(s)$ vectors on each non-independent bin. An edgeless stratum is returned directly.

For the remaining strata apply Lemma 4.17 with the displayed target and fixed surplus ϵ_{loc} . All local inputs use the supplied moment-derived vectors and degree bound D , and each local procedure constructs its own layers. The specified-target sufficient-cardinality exponent is $1 - a_0 + \beta G(L^*(s)) + e/8 + \beta e/16 + o(1)$. The loss function G is continuous, nondecreasing, and Lipschitz: it is a finite minimum of nondecreasing Lipschitz polynomial branches, composed with maxima and positive parts. The map L^* also has a fixed Lipschitz constant. Thus all bin and outward-parameter losses, as well as integer and logarithmic factors, fit an additional $e/16$ for large n . The total overhead is strictly below the reserve in Eq. (18), since

$$e/8 + \beta e/16 + e/16 = (3 + \beta)e/16 < e/4.$$

The equality collects the three terms, and the inequality uses $\beta < 1$. These fixed choices leave the later $e/2$ nonclosure reserve unchanged. Fresh conditional amplification makes the failure bound simultaneous over all calls. Only independently verified sets of the required size are submitted as progress. The local vectors justify the cardinality cap; the original SoS vectors are still used for main propagation. $\square$

4.6 Endpoint states and their cardinality caps

The state cap records the endpoint cardinality bound supplied by root-stratum rounding.

Definition 4.19 (State-cap function). *Let L^* be the loss in Lemma 4.5, and let G be the local loss from Definition 4.13. Define*

$$M(s) := \min\{1 + e, 1 - a_0 + \beta G(L^*(s)) + e\}, \quad m_{\text{init}} := \beta + e.$$

An endpoint state comprises a nonempty deterministic set X in one padded root-correlation bin B^+ , an actual least-indexed representative with correlation s , and a nonnegative stored threshold λ . The analytic covered-state invariant at depth j is

$$\Pr[\max_{x \in X} g \cdot w_x \geq (\lambda - \epsilon_n)t] \geq \exp(-c_j \epsilon_n t^2),$$

for fixed constants c_j . The algorithm never tests this probability. It enumerates all candidate states, applies amplified root rounding to each candidate, and submits only verified independent sets of size at least n^{1-a_0} . A simultaneous good event $\mathcal{G}$ ensures the conclusions of all estimates and rounding calls; conditional amplification makes its complement arbitrarily inverse-polynomially small.

If rounding on X has not returned the target on $\mathcal{G}$, then Lemma 4.18 and $|X| \leq n$ give, uniformly on B^+ and for large n ,

$$|X| \leq \exp((M(u) - e/2)t^2/2) \quad (u \in B^+). \quad (19)$$

The local target shift costs $e/8$ in the cardinality exponent. The fixed Lipschitz constant of $G \circ L^*$ and the vanishing bin and rounding losses fit the remaining $e/2$ reserve. This is a consequence on $\mathcal{G}$, not a deterministic inference from a randomized failure.

Taking a union bound for the covered-state invariant gives $\lambda^2 \leq 2 \log |X|/t^2 + O_J(\epsilon_n)$. This explains the raw cap test: after rounding, discard a candidate if its raw threshold exceeds its counting cap by more than a fixed multiple of ϵ_n . A covered candidate cannot be discarded this way on $\mathcal{G}$. No progress claim is made for an uncovered discarded set.

Choose a certified lower bound $M_B^- \leq \inf_{u \in B^+} M(u)$ with vanishing error, and a downward-rounded square root $r_B \leq \sqrt{M_B^-}$ with vanishing error. Clip the stored threshold to $\min\{\lambda, r_B\}$ only after the raw test. Then $\lambda^2 \leq M(u)$ for every member of the bin. For the initial neighborhood use m_{init} for this clipping cap.

The following map expresses one propagation step in terms of the parent threshold, correlations, and cardinality cap.

Definition 4.20 (Controlled child-threshold map). *For parent and child correlations s, τ put*

$$y(s, \tau) := \frac{s + 2\tau}{\sqrt{3}r(s)}, \quad A(s, \tau) := \frac{1 + 2s\tau}{2r(s)}.$$

For a control c and a cardinality cap $m \geq \lambda^2$ define

$$F_{c,m}(s, \lambda, \tau) := \frac{A(s, \tau)\lambda + Q_c(y(s, \tau), \sqrt{m - \lambda^2})}{r(\tau)}.$$

The first propagation uses $(c, m) = (c_1, m_{\text{init}})$ and later ones use $(c(s), M(s))$. In the latter case write $F(s, \lambda, \tau)$.

All exact retained pairs lie in $\mathcal{I}^2$, where $\mathcal{I} := [-1/2 - 10^{-11}, 3/4 + 10^{-11}]$, with $s + \tau \leq 1/2 + 10^{-11}$. Indeed the moment floor and Lemma 4.1 give the lower and sum constraints. Lemma 5.5 gives $|y| < 5/8$, and the edge identity implies $|\tau| \leq \sqrt{139}/16 < 3/4$. All normalization denominators are therefore bounded below. Implemented directions satisfy the same statements with vanishing collars, eventually inside the displayed enlarged domain. The actual retention allowance η is strictly below the certificate allowance e .

4.7 State propagation and survival before clipping

We implement this map by enumerating retained pairs and storing a certified minimum for each child bin.

Lemma 4.21 (Controlled state propagation). *Suppose a covered parent state has every required outgoing packing into the next layer W . Use a single grid control selected from its representative before choosing a child. At depth 1 suppose $\lambda \geq \sqrt{3b} - o(1)$ and $|X| \leq D$; at later depths suppose $\lambda \geq U(s) + 1/800000$ and Eq. (19) holds. Procedure PROPAGATEENDPOINTSTATE (Algorithm 4) returns finitely many deterministic candidates, one of which is covered at depth $j + 1$. Its raw threshold is*

$$\lambda_{B'} = F_{c,m(s_x)}(s_x, \lambda, \tau_v) + o(1)$$

for a specified minimum-attaining witness pair (x, v) in that candidate. Here $m(u) = m_{\text{init}}$ at depth 1 and $m(u) = M(u)$ later. The representative correlations differ from those of the witness by $o(1)$. Every error is uniform over the fixed depth and fits the certificate collars.

Proof. Apply Lemma 4.12 with $T := t$, $\theta := \beta$, $k := 1/2$, and $v := \sqrt{3}/2$. The main instances of Lemmas 4.10 and 4.11 supply its fiber hypothesis. At depth 1 use $m_x := m_{\text{init}}$ and $\epsilon := e$. At later depths use $m_x := M(s_x)$ and $\epsilon := e/2$, as guaranteed by Eq. (19). The clipping invariant makes $m_x - \lambda^2$ nonnegative, and Lemma 5.5 supplies $A > 0$ and the normalization bounds. The pair threshold is exactly $F_{c,m(s_x)}(s_x, \lambda, \tau_v)$. We check the implementation and deterministic witness selection explicitly.

Algorithm 4 Propagate an endpoint state with a parent-selected control

```

1: procedure PROPAGATEENDPOINTSTATE( $(X, B, s, \lambda), W, c, m$ ) ▷ Lemma 4.21
2:    $\mathcal{P} \leftarrow \emptyset$  and  $\mathcal{S} \leftarrow \emptyset$ .
3:   for all  $x \in X$  and  $v \in W \cap N_G(x)$  do
4:     Compute certified  $s_x, \tau_v, y_{xv}$  and the retention residual.
5:     if the retention interval meets  $(-\infty, 0]$  then
6:       Add  $(x, v)$  to every child bin met by its correlation interval.
7:     end if
8:   end for
9:   for all nonempty child pair bins  $\mathcal{C}_{B'}$  do
10:    Compute intervals  $[F_p^-, F_p^+]$  for all  $p \in \mathcal{C}_{B'}$ .
11:     $p_{B'} \leftarrow \text{lexmin argmin}_{p \in \mathcal{C}_{B'}} F_p^-$ .
12:     $\lambda_{B'} \leftarrow \max\{0, F_{p_{B'}}^- - C_P \varepsilon_n\}$ .
13:     $Y_{B'} \leftarrow \{v : (x, v) \in \mathcal{C}_{B'}\}$ ; choose its least-indexed representative.
14:    Append  $(Y_{B'}, B', \lambda_{B'}, p_{B'})$  to  $\mathcal{S}$ .
15:   end for
16:   return  $\mathcal{S}$ .
17: end procedure

```

For implemented directions, set $c_{xv} := \hat{v}_x \cdot \hat{v}_v$, $r_{xv} := \sqrt{1 - c_{xv}^2}$, and decompose $\hat{e}_{xv} = yq_x + u_{xv}$. The exact coefficients are

$$\hat{\tau} = c_{xv}s_x + r_{xv}y r(s_x), \quad \hat{A} = -c_{xv}r(s_x) + r_{xv}y s_x.$$

They differ from the nominal formulas by $O(\rho_n)$, and the coefficient of u_{xv} is $r_{xv} = \sqrt{3}/2 + O(\rho_n)$. Thus the approximate moment system is not treated as exact.

The shared proof bounds bad outer successes by $\exp(-et^2/8)$ for large n , including $\xi_x \leq h$ and $\xi_x = o(1)$. The tilted-projection error is $O(t^{-1/2} + \rho_n) = o(\varepsilon_n)$; the remaining geometric and scalar errors are $O(\varepsilon_n)$. Thus it gives the next covered-state invariant with the required uniform error scale.

All retained pairs are determined before the auxiliary Gaussian draw. Enumerating every retained graph neighbor only enlarges the family. There are $\exp(O(t))$ child bins; hence one deterministic bin carries at least $\exp(-c_{j+1}\varepsilon_n t^2)$ probability, for an effective fixed c_{j+1} . For that bin compute outward intervals for every nominal F , including the implemented-geometry and grid-control errors, to total width $o(\varepsilon_n)$. Choose the lexicographic minimizer of their lower endpoints. Subtract $C_P \varepsilon_n$ with a sufficiently large fixed C_P . This is a finite rational minimum, not a decision of an exact-real minimum. Every pair threshold dominates the stored threshold, so the distinct child vertices retain the cover guarantee.

Each enumerated pair satisfies the certificate retention condition for the exact profile for large n : interval error and the grid-control error fit the fixed gap $e - \eta = 10^{-9}$. The floor and triangle collars are also smaller than 10^{-11} . The certificate applies to the selected pair, as well as to every other retained pair above the profile. At later steps all members of the parent bin are above their own U , since the fixed surplus dominates bin variation. At the first step use the initial-entry bound and monotonicity in λ . These same bounds make the maximum with zero inactive on the covered chain. The witness and representative share padded bins, proving the comparison in the statement. $\square$

The raw counting test must preserve a covered child before its threshold is clipped to the state cap.

Lemma 4.22 (Survival and downward clipping). *On $\mathcal{G}$, a covered raw child either yields the target independent set or survives the raw cap test. In the latter case clipping costs $o(1)$ and its stored threshold has square at most $M(u)$ for every member of its padded bin. The selected transition and all finite-precision errors therefore transfer to the clipped state with only a uniform $o(1)$ loss.*

Proof. The covered child's Gaussian union bound and Eq. (19) give $\lambda_{B'}^2 \leq \inf_{u \in (B')^+} M(u) + o(1)$. This bound is obtained for the raw threshold, before clipping. Let M_B^+ be a certified upper bound for the cap on the padded bin, M_B^- a lower bound for its infimum, and r_B its downward-rounded square root. Their errors and bin oscillations tend to zero. Then

$$0 \leq \lambda_{B'}^2 - \min\{\lambda_{B'}, r_B\}^2 \leq (\lambda_{B'}^2 - M_B^+)_+ + (M_B^+ - M_B^-) + (M_B^- - r_B^2) = o(1).$$

The first term uses raw nonclosure, the second continuity of M and certified evaluation, and the third downward square-root precision. The square-root modulus also makes the unsquared loss $o(1)$. The clipped threshold preserves the cover and is below every bin cap. This reasoning is needed only for the covered candidate, not for all enumerated candidates. $\square$

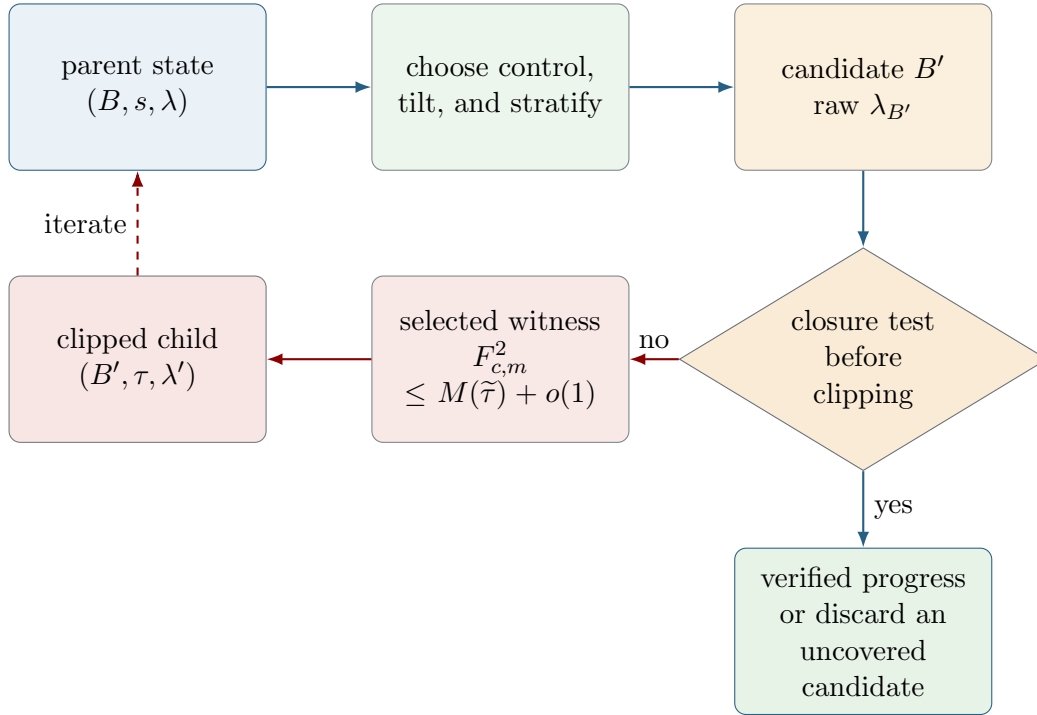


Figure 7: A continuing state transition with a parent-selected control. Raw closure is tested before clipping. The proof follows a covered child; other enumerated candidates may be discarded without a progress claim.

4.8 Initialization and amplified profile surplus

After constructing the menu layers, put $N := |S_J| = n - o(n)$ and apply the seed matching experiment inside S_J . Its selection and unmatched-tail exponents are b and $e/2$, so its cost is $b + e/2 = a_0 - e/2$. Use the general form of Lemma 4.8, for one seed marking level, with

$$P := n, \quad \tau := 1 - a_0, \quad c := a_0 - e/2, \quad \epsilon := e/4.$$

For sufficiently large n , its size hypothesis holds because $N = n - o(n) \geq n^{1-e/4} = P^{\tau+c+\epsilon}$. The marking cutoff is $Nn^{-e/8}$; the seed step leaves $N - o(N)$ unmarked vertices unless it returns a verified independent set of size at least $\lceil n^{1-a_0} \rceil$. The same marking argument therefore either returns progress or gives a root $a \in S_J$ whose neighborhood $X_1 := N_{G[S_J]}(a)$ covers at threshold $(\sqrt{3b} - o(1))t$. The algorithm may enumerate all roots; no test of a Gaussian cover probability is needed. Its neighbors belong to S_J and have every propagation packing into S_{J-1} . Moreover,

$$|X_1| \leq D = \exp(\beta t^2/2), \quad s_1 = -1/2 + o(1), \quad \lambda_1 = \sqrt{3b} - o(1). \quad (20)$$

The initial binning loses at most $\exp(O(t))$ in cover probability. The initial threshold lies below $\sqrt{m_{\text{init}}}$ by a fixed margin.

We measure each state's threshold relative to the profile to track progress across changing correlations.

Definition 4.23 (Normalized profile surplus). *Let U be the profile from Definition 5.4. Define $\mathcal{P}(s, \lambda) := \lambda/U(s) - 1$.*

The profile certificate amplifies this surplus at every continuing step, while the state cap keeps it bounded.

Corollary 4.24 (Fixed-depth termination above the profile). *On $\mathcal{G}$, the nested process with $J := 2048$ returns verified independent-set progress. A continuing covered chain cannot have 1800 transitions after the initialized depth-2 state.*

Proof. The initial inequality in Lemma 5.6, monotonicity of $F_{c_1, m_{\text{init}}}$ in its nonnegative threshold, and Lemma 4.22 give

$$\lambda_2 \geq U(s_2) + 1/400000, \quad \mathcal{P}(s_2, \lambda_2) \geq 1/400000. \quad (21)$$

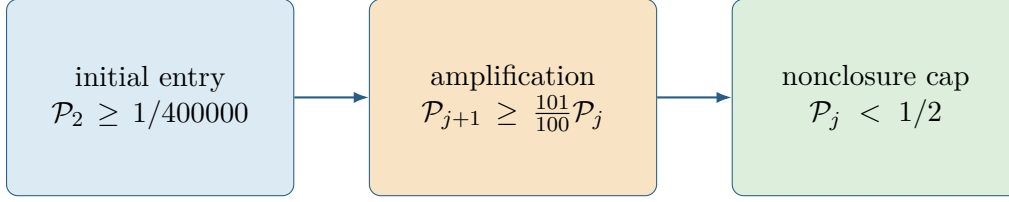
Here the certified numerator reserve is $1/200000$, the child radius is at most one, and all errors eventually cost less than half that reserve. The surplus bound then follows from $U(s_2) < 1$. For every later step the same lemma gives

$$\mathcal{P}(s_{j+1}, \lambda_{j+1}) \geq \frac{101}{100} \mathcal{P}(s_j, \lambda_j).$$

The fixed additive certificate margin 10^{-7} absorbs the uniform $o(1)$ losses in control approximation, bin representatives, moment data, threshold computation, and clipping. This induction also keeps every member of a parent bin above its profile by more than $1/800000$: the representative gap exceeds $3/1600000$ because $U > 3/4$, and the bin variation and representative errors eventually cost less than $1/1600000$. Since $U > 3/4$ and $\lambda^2 \leq 1 + e$ with $\sqrt{1+e} < 9/8$, every nonclosed surplus is less than $1/2$. But the binomial expansion gives $(101/100)^{100} > 2$, so 1800 further transitions would give

$$\mathcal{P} > \frac{2^{18}}{400000} = \frac{2048}{3125} > \frac{1}{2}.$$

This contradiction occurs before the 2048 layers are exhausted. $\square$



$\mathcal{P}(s, \lambda) := \lambda/U(s) - 1.$
 After 1800 further transitions the lower
 bound exceeds 1/2: progress is forced.

Figure 8: Profile termination. The initial transition enters above U ; every later transition amplifies normalized surplus. The nonclosure cap bounds that surplus, forcing progress at fixed depth.

Algorithm 5 Color a 3-colorable graph

```

1: procedure COLORTHREECOLORABLE( $G$ ) ▷ Theorem 4.25
2:   Set  $J \leftarrow 2048$ , initialize the sparse/dense wrapper, and fix all work and failure budgets.
3:   while the wrapper requests progress on  $H$  do
4:     if  $H$  is a dense request then
5:       Submit its deterministic dense-progress output to the wrapper.
6:     else
7:       Set the parameters using  $n \leftarrow |V(H)|$ ; handle  $n < N_0$  directly.
8:       Build the SoS data and deterministic control menu  $\mathcal{M}_n$ .
9:        $\mathcal{R} \leftarrow \text{CONSTRUCTNESTEDLAYERS}(H, J, C, \mathcal{M}_n, e/2).$  ▷ Algorithm 2
10:      if  $\mathcal{R}$  is a verified target independent set then
11:        Submit  $\mathcal{R}$  to the wrapper and continue to its next request.
12:      else if  $\mathcal{R} = \text{FAILURE}$  then
13:        If the sparse retry budget is exhausted, return FAILURE.
14:        Restart this sparse request with fresh randomness and continue.
15:      end if
16:       $(S_0, \dots, S_J) \leftarrow \mathcal{R}.$ 
17:      Run the seed marking experiment in  $S_J$ ; on verified progress, submit it and continue.
18:      Enumerate the depth-1 neighborhood states for every root in  $S_J$ .
19:      Perform the branch search continued below.
20:    end if
21:  end while
22:  return the verified coloring produced by the wrapper.
23: end procedure

```

4.9 The complete coloring algorithm

We complete the coloring algorithm by combining fixed-depth sparse progress with the sparse/dense reduction.

Theorem 4.25 (Main result, formal version of Theorem 1.1). *For every fixed $a' > 165227499/10^9$, there is a randomized polynomial-time algorithm coloring every 3-colorable n -vertex graph with $O(n^{a'})$ colors. In particular the bound is $O(n^{0.1652275})$.*

For each root the branch search performs the following finite operations. At depth j , put $W :=$

S_{J-j} , choose c_1, m_{init} if $j = 1$, and otherwise select the nearest-grid parent control with cap M . For every current state call $\mathcal{C} \leftarrow \text{PROPAGATEENDPOINTSTATE}((X, B, s, \lambda), W, c, m)$ (Algorithm 4). For every raw candidate call $\mathcal{K} \leftarrow \text{ROUNDROOTSTRATUM}$ (Lemma 4.18); submit verified target outputs, then test raw counting closure, and clip only continuing candidates. Retain every continuing state for the next depth. Stop after at most $J - 1$ transitions. A failed estimate or absent progress leads to a fresh sparse run within the fixed amplification budget. Exhausting that budget returns FAILURE, never an unverified output.

Proof. Fix $J := 2048$ before choosing the work and precision exponents. At fixed depth the menu and fine-bin counts are $\exp(O(t)) = n^{o(1)}$; each candidate examines at most nD neighbor pairs. Enumerating every root, every candidate, and every depth therefore takes at most n^{A_J} operations for an effective fixed A_J . Evaluation of the generated piecewise-affine profile takes polynomial bit time. The local routines have fixed layer budgets 16, 32, 128, 128, 128 and only invoke previously specified rounding routines. First fix the desired global failure exponent and all amplification, estimation, rejection, and restart caps, using their uniform inverse-polynomial success and acceptance bounds. Include every nested call and rejected proposal in a resulting ideal-work bound n^A . Only then choose C_0 using Lemma 4.2; the resulting ordinary-bit work is polynomial with a possibly larger exponent. Instantiating that lemma after this ideal-work bound also handles the variable but bounded Gaussian thresholds: replace t there by $\sqrt{q}t$; the rejection probabilities and anti-concentration bounds remain polynomial uniformly in q .

Choose C_0 and all conditional failure budgets so that $\Pr[\mathcal{G}] \geq 1 - n^{-60}$, and then choose N_0 large enough for every uniform error to fit the fixed reserves. The scalar square roots have a uniform square-root modulus even at the active hinge or a zero radicand; no unbounded derivative there is assumed in transferring the graph states. The normalized surplus has bounded derivatives on the compact state range because $U > 3/4$ and the fixed piecewise-affine slopes are bounded.

On $\mathcal{G}$, Lemma 4.8 either returns progress or supplies every required packing. Seed initialization and Lemma 4.21 then provide a covered child at every step unless rounding has already returned progress. Lemma 4.22 prevents deletion of that covered chain by a premature clipping or cap test. Corollary 4.24 rules out its continuation through the available depth. Hence a verified independent set of size $n^{1-a_0-o(1)}$ is found with the required uniform inverse-polynomial probability; our chosen target n^{1-a_0} is also allowed by the fixed peeling reserves and the buffered cap argument. Apply Lemma 4.4 to obtain $O(n^{a'})$ colors for every fixed $a' > a_0$. The finite range $n < N_0$ is colored directly and absorbed in the absolute constant. This proves the theorem. $\square$

5 Threshold-profile estimates

This section proves the scalar assertions needed for the local rounding chains and the threshold-profile method. Section 5.1 verifies the finite local chain recurrences. Section 5.2 defines the rational profiles and reduces their parameter margins to finite interval checks. Section 5.3 states the certificate inequalities for entry above the profile after the first propagation and for amplification at later transitions. Section 5.4 reduces the full threshold interval to explicit lower bounds. Section 5.5 justifies the whole-rectangle enclosures and the rules for covering the domain. Finally, Section 5.6 specifies the finite checks and gives their certified margins to complete the profile proof.

5.1 Exact certificates for the local chains

We first certify the local procedures of Theorem 4.16. Their controls are explicit functions of L and the parent correlation; no threshold-profile table is used. The respective scalar ranges for chains

0, 1, 2, 4, 5 are

$$\mathcal{I}_0 := [199/10000, 341/1000], \quad \mathcal{I}_1 := [29/1000, 67/200], \\ \mathcal{I}_2 := [159/1000, 67/200].$$

Put also

$$\mathcal{I}_4 := [2299/10000, 67/200], \quad \mathcal{I}_5 := [649/10000, 1401/10000].$$

These ranges contain all parameters used by the graph procedures, including the subordinate projected calls.

We fix one outward-rounded arithmetic system for constructing the profiles and verifying all scalar certificates.

Definition 5.1 (Exact interval arithmetic). *Put $S := 2^{46}$ and define*

$$\text{rd}(t) := \lfloor St \rfloor / S, \quad \text{ru}(t) := \lceil St \rceil / S, \quad [t] := [\text{rd}(t), \text{ru}(t)].$$

We also write $[t]_S := \text{rd}(t)$ and $[t]_S := \text{ru}(t)$. Write X^- and X^+ for the lower and upper endpoints of an interval X , respectively; both lie in $S^{-1}\mathbb{Z}$. For an interval expression E , the superscripts in E^- and E^+ refer to its endpoints after the prescribed interval evaluation. In particular, $(y^2)^-$ is the lower endpoint of the interval square, not the square of y^- . Enclose every rational input by $[t]$. Addition and subtraction use

$$X + Y = [X^- + Y^-, X^+ + Y^+], \quad X - Y = [X^- - Y^+, X^+ - Y^-].$$

Multiplication takes the minimum and maximum of the four endpoint products and rounds outward. Division uses the four endpoint quotients and requires $Y^- > 0$. For $\text{sq}(X)$, the lower unrounded endpoint is zero if $0 \in X$, and otherwise the smaller squared endpoint; the upper endpoint is the larger squared endpoint. Round both outward. For $\sqrt{X}$, require $X^- \geq 0$ and use monotonicity and outward rounding. At an endpoint $p/S \geq 0$, the lower root numerator is the integer q satisfying $q^2 \leq pS < (q+1)^2$. The upper numerator is q if $q^2 = pS$ and $q+1$ otherwise. Thus square roots require only exact integer operations.

Interval minimum and maximum act separately on the two endpoints. The hull of X, Y is $[\min\{X^-, Y^-\}, \max\{X^+, Y^+\}]$. Repeated occurrences retain their interval enclosures. All integer operations are exact; an invalid interval, denominator, radicand, or integer operation causes failure, never pruning or acceptance.

The local recurrences require uniform geometric margins so that projection and composition remain valid throughout each chain.

Lemma 5.2 (Uniform local parameter margins). *On the respective enlarged ranges, the controls in Definition 4.15 satisfy*

$$0 < x_i(L) < L, \quad q > 0, \quad H > 0, \quad b_0 > |z(s)|, \quad h(s) > 0, \quad \varrho(s)^2 > h(s)^2 + u.$$

Every reachable retained state has root correlation bounded away from -1 and 1 . The retained fiber norms are bounded away from zero, and $A, d_{L,s}(y) > 0$ on the reachable retained domain. All strict inequalities have fixed positive margins.

Proof. The identities $q + H^2 = x_i(L)$ and $z(s)^2 + h(s)^2 = H^2$ follow from the definitions. The tilt truncation gives $h(s)^2 \geq 111H^2/400$. Positivity of x_i , $L - x_i$, and the displayed composition reserve is checked by outward interval arithmetic on every loss interval. For the zero-tilt chain, put

$$Y_0^2 := 1 - \frac{b_0^2}{1 + H^2 + u}, \quad R_0^2 := k^2 + v^2 Y_0^2.$$

Retention implies $|y| \leq Y_0$. The edge formula $t = -ks + vr(s)y$ therefore gives $|t| \leq R_0$ by Cauchy–Schwarz. The interval checks prove $R_0 < 1$, $R_0^2 < k$, and that the entire seed correlation collar lies below R_0 . Thus $A = (k + st)/r(s) > 0$, even when the endpoint signs are opposite.

For chain 1 the interval check uses the conservative bound

$$Y_1^2 := 1 - \frac{(b_0 - 17H/20)^2}{1 + H^2 + u}, \quad R_1 := \max\{k + 10^{-8}, \sqrt{k^2 + v^2 Y_1^2}\} < 1.$$

Indeed $b_0 - zy \geq b_0 - 17H/20 > 0$ on $|y| \leq 1$, and retention implies $y^2 \leq Y_1^2$. This proves the normalization bound for every child and also includes the initial collar.

For completeness, a sharper analytic bound explains positivity of A in the tilted geometry, and supplies the radius used for chains 2, 4, and 5. Put $K := 1 + H^2 + u$ and

$$a_y := \sqrt{1 - b_0^2/K}, \quad R^2 := k^2 + v^2 a_y^2.$$

For $s \geq 0$ one has $z \geq 0$. Solving the retention quadratic gives

$$y_{\pm} = \frac{b_0 z \pm \sqrt{(K - z^2)(K - b_0^2)}}{K}, \quad -a_y \leq y \leq a_y + b_0 z/K.$$

For a negative child, $|t| \leq ks + vr(s)a_y \leq R$. For a nonnegative child, the bound $z \leq \chi L \sqrt{q}s/r(s)$ gives

$$t \leq -(k - \frac{2\chi L q}{(1+L)K})s + vr(s)a_y \leq R.$$

For chain 1, the rational interval checks on the 1224 loss intervals prove

$$k - \frac{2Lq}{(1+L)K} > \frac{19}{50}, \quad k - R^2 > \frac{9}{1000}, \quad 1 - R^2 > \frac{2}{25}.$$

These three margins can be reproduced with exact rational endpoint intervals on $[j/4000, (j+1)/4000]$, $116 \leq j < 1340$: evaluate the displayed rational formulas for x_1, q, H^2, k , then set $K := 1 + H^2 + u$, $b_0^2 := q/L$, $v^2 := 4L/(1+L)^2$, and $R^2 := 1 - v^2 b_0^2/K$. Use the four-endpoint product and positive-denominator quotient rules, retaining every repeated occurrence as an interval, and check that the lower endpoint of each of the three displayed margins exceeds its stated rational bound. Hence every actual retained state has $|s| \leq R$, with simultaneous sign reversal covering $s < 0$. It follows that

$$A \geq k\sqrt{1 - R^2} - vRa_y > 0,$$

because the difference of the squared terms is $k^2 - R^4 > 0$. For chain 2, on each interval $[j/8000, (j+1)/8000]$, $1272 \leq j < 2680$, the interval evaluation additionally checks

$$0 < a_y^2 < 1, \quad R < 1, \quad k - R^2 > 0, \quad k + 10^{-8} < R, \quad k - \frac{2Lq}{(1+L)K} > 0.$$

The same tests for chains 4 and 5 use intervals $[j/10000, (j+1)/10000]$, with $2299 \leq j < 3350$ and $649 \leq j < 1401$, respectively. Here each inequality means strict separation of the corresponding interval endpoints. The preceding argument then applies with this R , including the initial collar. Every tilted check independently requires a positive lower bound on A over every retained whole-cell entry that it uses, including its larger rectangular correlation enclosures.

Finally $b_s(y) > 0$ and retention give

$$b_s(y) \leq \sqrt{1 - y^2} \sqrt{1 + h(s)^2 + u} < \sqrt{1 - y^2} \sqrt{1 + \varrho(s)^2},$$

which proves $d_{L,s} > 0$. Base positivity and the same retention inequality bound $1 - y^2$ away from zero. All formulas are continuous on the resulting compact domain, including the tilt truncation seams. $\square$

The next certificate excludes every continuing local trajectory within the available layer budget.

Lemma 5.3 (Local-chain certificate). *Put $\delta_{\text{loc}} := 10^{-5}$. Use the exact controls and caps of Definition 4.15 and retain a pair when*

$$b_s(y)^2 \leq (1 + h(s)^2 + u)(1 - y^2).$$

Start at state depth 1 with

$$|s_1 + k| \leq 10^{-8}, \quad \lambda_1 \geq \sqrt{x_i(L)/L} - \delta_{\text{loc}}, \quad \lambda_1^2 \leq 1 + u.$$

The first propagation uses cap $1 + u$ and zero tilt except in chain 5, which uses its parent-selected tilt; later ones use the specified parent control and $m_i(L, s)$. Suppose every continuing transition satisfies

$$\lambda_{j+1} \geq \max\{0, F_{i,m}(L, s_j, \lambda_j, s_{j+1}) - \delta_{\text{loc}}\}, \quad \lambda_{j+1}^2 \leq m_i(L, s_{j+1}). \quad (22)$$

No such trajectory reaches state depths 10, 28, 37, 40, and 25 for chains 0, 1, 2, 4, and 5 on their respective enlarged ranges.

Proof. We specify the complete finite check; its only data are the rational constants and formulas in Definitions 4.13 and 4.15 and the present lemma. All arithmetic uses Definition 5.1.

For integers d, c, j , use the loss interval $[\lfloor j/d \rfloor_S, \lceil (j+1)/d \rceil_S]$ and correlation mesh $1/c$. The complete index ranges are

chain	d	c	loss indices j
0	4000	1000	$200 \leq j < 1364$
0	10000	3000	$199 \leq j < 501$
1	4000	1000	$116 \leq j < 1340$
2	8000	1500	$1272 \leq j < 2680$
4	10000	1500	$2299 \leq j < 3350$
5	10000	1500	$649 \leq j < 1401$

The two chain-0 ranges overlap on $[1/20, 501/10000]$ and together cover $\mathcal{I}_0$. Evaluate the controls, caps, and reachable-radius expressions by the interval operations just specified, retaining the separately computed interval $H^2 := x(L - x)/(1 + L)$ when forming $h := \sqrt{H^2 - z^2}$. The quantities H^2, Y_0^2, Y_1^2 retain their directly defined interval expressions. Every other squared computed quantity means its interval square. For chain 0 use $R_0 := \sqrt{k^2 + v^2 Y_0^2}$ from Lemma 5.2; for chain 1 use the displayed R_1 there, with its denominator $1 + H^2 + u$ evaluated as $1 + \text{sq}(H) + u$ after computing $H := \sqrt{H^2}$. Here sq denotes the interval squaring operation above. Chains 2, 4, and 5 instead use the sharper R and all five strict tests in Lemma 5.2, evaluating $a_y^2 := 1 - \text{sq}(b_0)/(1 + H^2 + u)$ and $R := \sqrt{\text{sq}(k) + \text{sq}(v)a_y^2}$. The zero-tilt check requires $0 < (Y_0^2)^- \leq (Y_0^2)^+ < 1$, $R_0^+ < 1$, $(k - R_0^2)^- > 0$, and $k^+ + \lceil 10^{-8} \rceil_S < R_0^-$. Chain 1 requires $(b_0 - 17H/20)^- > 0$, $0 < (Y_1^2)^- \leq (Y_1^2)^+ < 1$, and $R_1^+ < 1$. Every check requires $x^-, (L - x)^-, q^-, (H^2)^- > 0$. Each zero-tilt check additionally requires $((H/x)^2 - H^2 - u)^- > 0$.

Let R^+ be the upper endpoint of the corresponding radius and put $n := \lceil cR^+ \rceil$. The correlation cells are exactly

$$C_a := [\lfloor a/c \rfloor_S, \min\{\lceil (a+1)/c \rceil_S, R^+\}], \quad 0 \leq a < n.$$

Thus all rational grid boundaries and their outward collars are covered. For every cell store $r(C_a) = \sqrt{1 - \text{sq}(C_a)}$ and the upper endpoint M_a of its cap $m_i(L, C_a)$. In the tilted case also store its parent controls. Evaluate all minima and maxima in the controls and cap by interval minimum and maximum, including both branches at a seam.

For zero tilt, opposite signs give a valid lower bound for either actual orientation. If $s, t \geq 0$, then $|ks + t| \geq |ks - t|$, so every retained same-sign pair has a retained opposite-sign pair. Moreover $A_+ \geq A_-$ and $d_+ \leq d_-$; all other terms and the cap are unchanged. Thus the same-sign transition is at least the opposite-sign transition at every nonnegative threshold. This is a scalar relaxation, not a merger of signed graph bins. For every tilted check, simultaneous reversal $(s, t, z) \mapsto (-s, -t, -z)$ preserves the recurrence. Its later parent cells are nonnegative, and both child orientations $t \in -C_b$ and $t \in C_b$ are checked separately.

Here are the coefficient bounds on a complete parent-child rectangle. For zero tilt take $s \in C_a$, $t \in C_b$ as absolute correlations, and evaluate $y := (ks - t)/(vr(s))$. Prune only if $(y^2)^- > (Y_0^2)^+$. Otherwise require $(y^2)^- < 1$ and set

$$A^- := ((k - st)/r(s))^- , \quad B^- := (vb_0)^- , \quad f^+ := (v/(\varrho + H))^+ , \quad g^+ := (\sqrt{1 + \varrho^2})^+ , \\ w^+ := \lceil \sqrt{1 - (y^2)^-} \rceil_S , \quad e^+ := \lceil w^+ g^+ \rceil_S - b_0^- , \quad d^+ := \lceil f^+ e^+ \rceil_S .$$

Require $A^-, e^+, d^+ > 0$. The use of $(y^2)^-$ here is deliberate: an unpruned rectangle can also contain infeasible points with $y^2 > 1$, but w^+ still bounds the fiber norm at every retained point. Positivity of the actual d at retained points follows from Lemma 5.2; no positive interval lower bound for d on these larger zero-tilt rectangles is needed.

For a tilted parent and either signed child, evaluate $y := (t + ks)/(vr(s))$. With the parent's interval controls put $C := 1 + h^2 + u$ and compute outward intervals for

$$y_{\pm} := \frac{b_0 z \pm \sqrt{C(C + z^2 - b_0^2)}}{C + z^2} .$$

Require $y_-^- > -1$ and $y_+^+ < 1$. Intersect the geometric y interval with $[y_-^-, y_+^+]$; an empty intersection is pruned. Using this intersection, set $b := b_0 - zy$ and prune if $(b^2 - C(1 - y^2))^- > 0$. Otherwise evaluate

$$A := \frac{k + st}{r(s)} , \quad B := vb , \quad d := \frac{v}{\varrho + h} (\sqrt{1 - y^2} \sqrt{1 + \varrho^2} - b)$$

by interval arithmetic, require $A^-, d^- > 0$, and retain the endpoints A^-, B^- , and d^+ . The original parent and child intervals are not narrowed when evaluating A or their radii. For each tilted parent, including the seed control, also require $(\varrho^2 - h^2 - u)^- > 0$.

For a parent lower threshold $\lambda \geq 0$, parent cap upper bound M , parent control lower bound h^- , and child radius upper bound r^+ , the exact scalar update is

$$X := \max\{0, \lceil \sqrt{M - \lfloor \lambda^2 \rfloor_S} \rceil_S - h^-\} , \quad N := \lfloor A^- \lambda \rfloor_S + B^- - \lceil d^+ X \rceil_S ,$$

$$T := \begin{cases} 0, & N \leq 0, \\ \max\{0, \lfloor N/r^+ \rfloor_S - \lceil \delta_{\text{loc}} \rceil_S\}, & N > 0. \end{cases}$$

The radicand must be nonnegative. Dividing a negative lower numerator by an upper radius is not used. Monotonicity in the parent threshold proves that T is valid for every actual threshold above λ and below its actual cap.

To initialize, put $\epsilon_c := \lceil 10^{-8} \rceil_S$ and $C_* := [k^- - \epsilon_c, k^+ + \epsilon_c]$. Use the absolute seed C_* for zero tilt and chains 2, 4, and 5, and the signed seed $-C_*$ for chain 1. The two tilted seed orientations

are equivalent by simultaneous reversal, since both child signs are tested. For chain 5, evaluate the parent controls on the entire seed cell C_* using the same interval tilt formula as at later steps. All other chains use seed controls $z := 0$, $h := H$, $\varrho := H/x$. The seed cap upper bound is $(1+u)^+$, and its threshold lower bound is $\lambda_* := (\sqrt{x/L})^- - \lceil \delta_{\text{loc}} \rceil_S$, which must be positive. Apply the stated coefficient rules and scalar update from this seed to every child cell, including both child signs in the tilted check. The resulting envelope is at state depth 2.

At each step the lower bound of a child cell is the minimum of T over all incoming unpruned entries from present parent cells and all required orientations. A cell with no incoming entry is absent. A finite child lower bound λ is removed only if $\lfloor \lambda^2 \rfloor_S > M_b$. All other cells are retained, even when they contain spurious infeasible points. This initializes and then inductively preserves a representation of every continuous continuing trajectory. Allowing L to vary within one loss interval between steps only enlarges this representation.

Any invalid interval, failed positivity or domain requirement, or integer arithmetic error is a failure of the check, not a prune or a pass. For each loss interval require an empty envelope at or before the stated terminal depth for its chain. A nonempty envelope at the required terminal depth causes failure. Acceptance requires successful completion for every prescribed loss interval. The complete checks on $[1/20, 99/1000]$ and $[99/1000, 341/1000]$ give empty zero-tilt envelopes by state depths 8 and 9, respectively. These ranges share their boundary and cover the chain-0 domain above $1/20$. The chain-1 envelopes on $[29/1000, 99/1000]$ and $[99/1000, 67/200]$ are empty by state depths 14 and 28, respectively. These ranges also share their boundary and cover $\mathcal{I}_1$. The additional chain-0 envelopes on $[199/10000, 501/10000]$ are empty by depth 9. Every chain-2 envelope is empty by depth 37. Every chain-4 envelope is empty by depth 40, and every chain-5 envelope is empty by depth 25. All prescribed intervals are included. This proves the asserted termination. $\square$

The complete checks satisfy every strict parameter test and give the empty terminal envelopes stated above. The rational data, interval operations, pruning rules, and envelope recursion specify these checks without additional input. Acceptance requires all strict domain tests and all terminal envelopes to pass; a partial run is not a certificate. These finite scalar checks combine with the matching, specified-target rounding, and ordinary-bit transfer in Theorem 4.16.

5.2 Rational profiles and uniform margins

Use $a_0 := 165227499/10^9$, $\beta := 1 - 2a_0$, $e := 10^{-8}$, $b := a_0 - e$, and $\gamma := 1/100$. Define $\mathcal{I} := [-1/2 - 10^{-11}, 3/4 + 10^{-11}]$. The controls are

$$q(s) := b - z(s)^2 - h(s)^2, \quad \varrho(s) := \beta h(s)/b.$$

The special first control is

$$z_1 := -\frac{7400728992}{10^{12}}, \quad h_1 := \frac{101675936082}{10^{12}}, \quad q_1 := b - z_1^2 - h_1^2, \quad \varrho_1 := \beta h_1/b.$$

All constants, controls, and caps in this section use the actual progress exponent a_0 directly. Its gap below the displayed exponent 0.1652275 is exactly 10^{-9} ; no perturbation from a different certified exponent is needed.

We specify the profile by a short control vector and a fixed recurrence. The recurrence constructs rational data; the subsequent whole-domain checks certify the resulting functions.

The control vector. Let $c_0 < \dots < c_{46}$ be the increasing enumeration of

$$\mathcal{C} := \{-1/2 + j/32 : 0 \leq j \leq 40\} \cup \{j/10000 : j \in \{-100, -48, -24, 0, 24, 48, 100\}\}.$$

For a vector v , write $v_{i:j} := (v_i, \dots, v_j)$. Define the integer vector ζ by

$$\begin{aligned}\zeta_{0:6} &:= (-2151178, 25853, 1366003, 1543903, 1595680, 1896940, 2112057), \\ \zeta_{7:13} &:= (2304005, 2461107, 2610609, 2431665, 2003704, 1582057, 1165335), \\ \zeta_{14:20} &:= (752229, 346723, 68136, 0, -31445, 0, 31445), \\ \zeta_{21:27} &:= (0, -68136, -341480, -752229, -1165355, -1582057, -2003704), \\ \zeta_{28:34} &:= (-2431665, -2480720, -2028076, -1547119, -1326279, -1068237, -839185), \\ \zeta_{35:41} &:= (-627322, -432567, 0, 520065, 1159880, 2098382, 2315262), \\ \zeta_{42:46} &:= (2806125, 3559277, 5232688, 5300117, 4952181).\end{aligned}$$

Let $\hat{z}$ interpolate $\hat{z}(c_i) := \zeta_i/10^8$ affinely. Put $H := 103567175/10^9$ and $Q(x) := 10^{-12} \lceil 10^{12}x \rceil$.

The knot set. The knots include a uniform grid and a finer central grid. We also place knots on both sides of the cap breakpoints. To specify these without numerical root data, put

$$\mathcal{L} := \{0, 1/50, 3/100, 13/200, 7/50, 4/25, 23/100, 1/3\}.$$

On each open interval between consecutive elements of $\mathcal{L}$, take the polynomial expressions for $L, g_0, g_1, g_2, g_4, g_5, a_0/\beta$, where

$$\begin{aligned}g_0(L) &:= \frac{61}{100} \max\{L, 1/50\} + \frac{13}{25} \max\{L, 1/50\}^2, \\ g_1(L) &:= \frac{19}{2000} + \frac{221}{500} \max\{L, 3/100\} + \frac{427}{500} \max\{L, 3/100\}^2, \\ g_2(L) &:= \frac{17}{2000} + \frac{221}{500} \max\{L, 4/25\} + \frac{427}{500} \max\{L, 4/25\}^2, \\ g_4(L) &:= \frac{1}{125} + \frac{221}{500} \max\{L, 23/100\} + \frac{427}{500} \max\{L, 23/100\}^2, \\ g_5(L) &:= \frac{14}{25} \max\{L, 13/200\} + \frac{1}{2} \max\{L, 13/200\}^2 + (L - 7/50)_+.\end{aligned}$$

Let $\mathcal{R}$ contain $\mathcal{L}$ and every root in the corresponding open interval of a nonzero difference of two of these polynomials. Only linear and quadratic equations occur. For each $\ell \in \mathcal{R}$ and each $x \in \{(3\ell - 1)/(6\ell + 2), (1 - 3\ell)/4\}$, put $n(x) := \lfloor 10^{10}x + 1/2 \rfloor$. Let $\mathcal{B}$ consist of $(n(x) + j)/10^{10}$ for $j \in \{-1, 0, 1\}$ that lie strictly between $-1/2$ and $3/4$. Define

$$\mathcal{K} := \mathcal{C} \cup \mathcal{B} \cup \{-1/2 + j/800 : 0 \leq j \leq 1000\} \cup \{j/5000 : |j| \leq 50\} \cup \{j/20000 : |j| \leq 100\}.$$

All indices in these sets are integers. Remove duplicates and write $\mathcal{K} = \{s_0 < \dots < s_{N-1}\}$. This prescription gives $N = 1420$. The rounding of the quadratic roots is exact: enclose each square root between consecutive multiples of 2^{-180} and require the two bounds to give the same integer $n(x)$; use its exact rational value if the square root is rational. Every such test succeeds. At a knot put

$$\bar{z}_i := Q(\hat{z}(s_i)), \quad \bar{h}_i := Q(\sqrt{H^2 - \bar{z}_i^2}).$$

These two evaluations of Q use exact rational arithmetic and integer square roots, before passing their rational outputs to the interval arithmetic below.

The threshold recurrence. Use Definition 5.1 for every operation in this paragraph. Enclose every rational input, including $s_i, \bar{z}_i, \bar{h}_i$, and let a superscript $-$ or $+$ denote an interval endpoint. For an interval X put $\mathcal{Q}(X) := Q(X^+)$. Evaluate $\beta := 1 - 2a_0$, $b := a_0 - e$, and $\kappa := \sqrt{3}/2$ in that order. For every parent i precompute

$$r_i := \sqrt{1 - s_i^2}, \quad q_i := b - \bar{z}_i^2 - \bar{h}_i^2, \quad \rho_i := \beta \bar{h}_i / b, \quad m_i := M(s_i),$$

$$b_i := \sqrt{3q_i}, \quad v_i := \sqrt{\beta + \rho_i^2}, \quad w_i := \sqrt{m_i}, \quad k_i := \sqrt{m_i - \bar{h}_i^2}, \quad \delta_i := 3 \cdot 10^{-7} + \min\{10^{-6}, s_i^2/100\}.$$

For m_i use the cap evaluation order specified in Section 5.5. For a pair i, j put $y_{ij} := (s_i + 2s_j)/(\sqrt{3}r_i)$. Discard the pair if $(s_i + s_j)^- > [1/2 + 10^{-11}]^+$, if $y_{ij}^- > [6251/10000]^+$, or if $y_{ij}^+ < -[6251/10000]^+$. For each remaining pair put $b_{ij} := b_i - \bar{z}_i y_{ij}$ and $l_{ij} := 1 - y_{ij}^2$. Discard it also if

$$(b_{ij}^2 - (\beta + \bar{h}_i^2 + e)l_{ij})^- > [10^{-6}]^+.$$

Let J_i be the remaining child indices, and for $j \in J_i$ precompute

$$A_{ij} := \frac{1 + 2s_i s_j}{2r_i}, \quad B_{ij} := \kappa b_{ij}, \quad d_{ij} := \frac{\kappa(\sqrt{l_{ij}}v_i - b_{ij})}{\rho_i + \bar{h}_i},$$

$$V_{ij} := A_{ij}^2 + d_{ij}^2, \quad K_{ij} := B_{ij} + d_{ij}\bar{h}_i.$$

Each J_i is nonempty. Set $u_i^{(0)} := 19/25$ for every i . For $k = 0, \dots, 99$, compute the following intervals for every $j \in J_i$; the right side uses only the preceding vector $u^{(k)}$. Suppress the indices on $A, B, d, V, K, m, w, h, \delta$ and put

$$R := r_j[u_j^{(k)}], \quad C := R + \delta - K,$$

$$X_a := \frac{AC + d\sqrt{\max\{0, Vm - C^2\}}}{V}, \quad X_b := \frac{R + \delta - B}{A}, \quad T := R + \delta - Ak_i - B.$$

Set $X := X_b$ if $T^- \geq 0$, set $X := X_a$ if $T^+ < 0$, and otherwise set $X := \max\{X_a, X_b\}$. Next put

$$\xi := \sqrt{\max\{10^{-18}, m - X^2\}}, \quad E := \frac{(1 + \gamma)R}{\max\{X, 10^{-9}\}} - A,$$

$$C_* := K + \gamma R - \delta, \quad D := C_*^2/m - d^2, \quad P := dX/\xi - E.$$

Set $Y := (1 + \gamma)R/(A + \sqrt{D})$ if $D^- > 0$, and set $Y := [2]$ otherwise. Set $Z := X$ if $P^- \geq 0$, set $Z := Y$ if $P^+ < 0$, and otherwise set $Z := \max\{X, Y\}$. Finally put

$$T_{ij}^{(k)} := \max\{Z, (1 + \gamma)Rw/(B + \gamma R + Aw - \delta)\},$$

$$u_i^{(k+1)} := \mathcal{Q}(\max_{j \in J_i} T_{ij}^{(k)}). \tag{23}$$

All interval minima and maxima act on endpoints separately. Every square uses the interval square operation. Evaluate the expressions in the displayed order without algebraic rearrangement. The construction stops after exactly 100 simultaneous updates; it has no convergence test or unspecified optimization step.

Affine interpolation extends the generated knot data to the full correlation domain.

Definition 5.4 (Exact rational profile). *Let $\mathcal{K}$ and $\bar{z}_i, \bar{h}_i$ be constructed above, and let $u_i^{(100)}$ be given by Eq. (23). Put*

$$U(s_i) := u_i^{(100)}, \quad z(s_i) := \bar{z}_i, \quad h(s_i) := \bar{h}_i.$$

Interpolate the three functions affinely between consecutive knots and extend the first and last affine pieces to the endpoint collars of $\mathcal{I}$. These prescriptions define continuous piecewise-affine functions with finite fixed Lipschitz constants. The separate first control is (z_1, h_1) specified at the beginning of this subsection.

The construction completes with all interval operations defined. The certificate below checks its output on whole intervals and rectangles, including the complete threshold range. Thus its proof does not rely on treating the construction grid as a sampling argument.

We first verify the geometric margins needed for projection and composition.

Lemma 5.5 (Certified parameter margins). *For every $s \in \mathcal{I}$, the profiles and controls satisfy*

$$\frac{3}{4} < U(s) < \frac{9}{10}, \quad q(s) > 0, \quad h(s) > \frac{7}{100}, \quad U(s)^2 + h(s)^2 < M(s), \quad \varrho(s)^2 > h(s)^2 + e.$$

The first control also has $q_1 > 0$, $h_1 > 7/100$, and $\varrho_1^2 > h_1^2 + e$. The base threshold $b_c(y)$ is positive for all $|y| \leq 1$. Every retained fiber, with either the parent control or the special first control and allowance e , has $|y| < 5/8$. Both $A(s, \tau)$ and $d_c(y)$ are positive on the certified retained domain. All these statements have fixed positive reserves where a strict inequality is asserted.

Proof. The finite interval checks are detailed in Section 5.6. We give the reduction they verify. On a whole interpolation piece enclose U, z, h and their affine slopes, then evaluate $q = b - z^2 - h^2$ and $\varrho = \beta h/b$ by outward-rounded arithmetic. The interval check verifies each displayed inequality over the full piece, including every knot seam and endpoint collar. For base positivity it checks $\sqrt{3q} > |z|$.

Write $v := \beta + h^2 + e$, $b_0 := \sqrt{3q}$, and $d_0 := v + z^2$. The retention condition is the quadratic inequality

$$(v + z^2)y^2 - 2b_0zy + 3q - v \leq 0.$$

Its two roots are

$$y_{\pm} := \frac{b_0z \pm \sqrt{v(d_0 - 3q)}}{d_0}.$$

The verifier checks that these are real and lie strictly between $-5/8$ and $5/8$, uniformly on each whole interpolation piece. It repeats this calculation for the special first control. This proves the retained-range claim. Next $1 + 2s\tau > 0$ on $\mathcal{I}^2$ and on the larger prescribed first-parent collar with $\tau \in \mathcal{I}$, which proves $A > 0$. Retention and $\varrho^2 > h^2 + e$ give

$$(1 - y^2)(\beta + \varrho^2) > b_c(y)^2.$$

Together with base positivity this proves $d_c(y) > 0$. Finally the edge formula and Cauchy–Schwarz give $|\tau| \leq \sqrt{1 + 3(5/8)^2}/2 = \sqrt{139}/16 < 3/4$. $\square$

5.3 Initial entry and uniform amplification

The two certificate inequalities place the initial state above the profile and amplify the surplus at every later transition.

Lemma 5.6 (Profile certificate). *Let U, z, h be defined in Definition 5.4, let M be the cap from Definition 4.19, and let $F_{c,m}$ be the map from Definition 4.20. Put $\delta_{\text{prof}} := 10^{-7}$. For $s, \tau \in \mathcal{I}$ satisfying*

$$s + \tau \leq 1/2 + 10^{-11}, \quad b_{c(s)}(y(s, \tau))^2 \leq (\beta + h(s)^2 + e)(1 - y(s, \tau)^2), \quad (24)$$

and every $U(s) \leq \lambda \leq \sqrt{M(s)}$, one has

$$\frac{F(s, \lambda, \tau)}{U(\tau)} - 1 \geq \frac{101}{100} \left(\frac{\lambda}{U(s)} - 1 \right) + \delta_{\text{prof}}. \quad (25)$$

For the initial control, put $\lambda_0 := \sqrt{3b} - e$ and $m_{\text{init}} := \beta + e$. For $|s + 1/2| \leq 10^{-10}$ and $\tau \in \mathcal{I}$ satisfying the triangle and retention conditions with control c_1 , one also has

$$r(\tau)(F_{c_1, m_{\text{init}}}(s, \lambda_0, \tau) - U(\tau)) \geq \frac{1}{200000}. \quad (26)$$

Proof. Section 5.4 reduces the full λ interval to two explicit bounds on each correlation pair. Section 5.5 proves that the verifier encloses those bounds on every rectangle and exhausts the domain. The executed integer-arithmetic checks in Section 5.6 prove every required lower bound. $\square$

5.4 Eliminating the complete threshold interval

Fix a retained pair and abbreviate

$$m := M(s), \quad U := U(s), \quad V := U(\tau), \quad R := r(\tau)V.$$

Put $B := \sqrt{3}b_c(y)/2$, $d := d_c(y)$, and $A := A(s, \tau)$. Define $E := (1 + \gamma)R/U - A$ and $C := B + dh + \gamma R$. The amplified numerator residual on the active branch is

$$H(\lambda) := C - E\lambda - d\sqrt{m - \lambda^2}.$$

Its derivative $-E + d\lambda/\sqrt{m - \lambda^2}$ is increasing. Lemma 5.5 ensures $U < \sqrt{m - h^2}$. If the derivative at U is nonnegative on the whole rectangle, then the active residual is at least

$$G_0 := AU + B - d(\sqrt{m - U^2} - h) - R.$$

Otherwise the Cauchy–Schwarz inequality gives the unconditional lower bound

$$G_* := C - \sqrt{m}\sqrt{E^2 + d^2}.$$

No stationary-point location is assumed in this second bound. The inactive residual is affine and agrees with the active residual at the hinge. Its minimum on the inactive interval is therefore bounded below by the same active lower bound or the full-cap value

$$G_{\text{top}} := B + \gamma R - E\sqrt{m}.$$

Consequently it suffices to prove $G_0, G_{\text{top}} \geq \delta_{\text{prof}}$ in the derivative-positive case, or $G_*, G_{\text{top}} \geq \delta_{\text{prof}}$ otherwise. Since $0 < R < 1$, division by R proves Eq. (25) with the stated additive margin.

For the seed, evaluate $A\lambda_0 + B - d(\sqrt{m_{\text{init}} - \lambda_0^2} - h_1) - R$. Its active formula is valid: $h_1^2 < b$ and $\beta + e - 4b = 1 - 6a_0 + 5e > 0$, while $\lambda_0^2 < 3b$. These exact inequalities imply $m_{\text{init}} - \lambda_0^2 > h_1^2$.

5.5 Whole-rectangle bounds and exhaustive coverage

The checks use the exact interval arithmetic of Definition 5.1. The acceptance rules are specified in Section 5.6. We first prove the enclosure and coverage principles that justify those rules.

The evaluation rules in Section 5.6 enclose each affine piece and its one-sided derivative, taking hulls over all intersected pieces and extending the endpoint pieces through the collars. For the cap, split at the four pieces of L^* separated by $-1/2, 0, 1/4$. Write $L := L^*(s)$ and set

$$\begin{aligned} w_0 &:= \max\{L, 1/50\}, & w_1 &:= \max\{L, 3/100\}, & w_2 &:= \max\{L, 4/25\}, \\ w_4 &:= \max\{L, 23/100\}, & w_5 &:= \max\{L, 13/200\}. \end{aligned}$$

Evaluate G from Definition 4.13 using $g_i := x_i(w_i)$ for $i \in \{0, 1, 2, 4\}$ and $g_5 := x_5(w_5) + (L - 7/50)_+$. In the recorded evaluation order, first form $\min\{L, \min\{g_0, \min\{g_1, g_2\}\}\}$, then take its minimum with g_5 , and finally with g_4 . The cap is

$$M(s) := \min\{1 + e, 1 - a_0 + \beta G(L^*(s)) + e\}.$$

All five costs use the exact rational coefficients in that definition. The minimum and maximum rules below enclose all possible active branches; their derivative hulls include every clipping point, quadratic-cost intersection, and global-cap switch. Continuity across these seams justifies the centered bound. The differentiable expressions have positive denominators: in particular $m - U^2 > h^2 > 0$ and $d > 0$.

A centered value and uniform derivative bounds give a lower bound on an entire rectangle, including knot and cap seams.

Lemma 5.7 (Rectangle lower bound). *Let f be continuous and piecewise continuously differentiable on a rectangle, and let c be a point in it. Let r_s, r_τ be the maximum coordinate distances from c to the rectangle's sides. Suppose the one-sided partial derivatives have interval bounds D_s, D_τ throughout the rectangle and $L_c \leq f(c)$. Then*

$$\inf f \geq L_c - r_s \sup |D_s| - r_\tau \sup |D_\tau|.$$

Proof. Integrate the derivative along the segment from c to any point of the rectangle, subdividing at its finitely many knot or cap crossings. Continuity makes the intermediate endpoint terms cancel. The derivative bounds control each integral and give the stated inequality. Segments lying on a seam follow by continuity from neighboring segments. $\square$

The program takes the larger of the direct interval lower bound and the bound of Lemma 5.7. If it uses G_0 , the derivative sign at U is certified on the entire rectangle. The midpoint expression uses that same branch of the bound, rather than selecting a different formula at the midpoint.

For domain pruning,

$$\partial_\tau y = \frac{2}{\sqrt{3}r(s)} > 0, \quad \partial_s y = \frac{1 + 2s\tau}{\sqrt{3}r(s)^3} > 0.$$

The positivity checks include the larger first-parent collar, so opposite corners enclose the fiber range on every main or seed rectangle. The program intersects the natural interval value of y with this corner enclosure, without narrowing its derivative intervals. The pruning tests in Section 5.6 therefore remove only rectangles disjoint from the retained domain. Its closed initial intervals and closed bisections include every seam and boundary. Every surviving rectangle must pass the required whole-rectangle bounds; an unresolved rectangle causes failure.

5.6 Verification of the finite certificate

We specify the finite checks independently of a programming language. Their inputs are the generated rational data in Definition 5.4, the first control, and the formulas in this section. The fixed recurrence specifies all inputs without a numerical search.

Arithmetic and expression evaluation. Use Definition 5.1 for all interval operations. Evaluate $\beta = 1 - 2a_0$, $b = a_0 - e$, $\sqrt{3}$, $\kappa = \sqrt{3}/2$, and β/b from these enclosures, and then evaluate the displayed expressions in their indicated order. Repeated occurrences retain their interval enclosures; they are not treated as independent numerical samples.

For derivative bounds, carry a triple $(f, D_s f, D_\tau f)$ of intervals with each expression. Constants have zero derivatives, and the two input coordinates have derivatives $(1, 0)$ and $(0, 1)$. For either coordinate $i \in \{s, \tau\}$ use

$$\begin{aligned} D_i(fg) &= (D_i f)g + f(D_i g), & D_i(f/g) &= ((D_i f)g - f(D_i g))/g^2, \\ D_i(f^2) &= 2fD_i f, & D_i\sqrt{f} &= (D_i f)/(2\sqrt{f}), \end{aligned}$$

with the interval operations just defined. For intervals $[u_-, u_+]$ and $[v_-, v_+]$, their minimum has value enclosure $[\min\{u_-, v_-\}, \min\{u_+, v_+\}]$; use the two maxima for a maximum. Select a derivative branch only under strict separation of the complete value intervals; otherwise hull both derivative intervals.

At a profile knot x_i let $X_i := [x_i]$ and let V_i be the outward enclosure of one of its three ordinates. Precompute $D_i := (V_{i+1} - V_i)/(X_{i+1} - X_i)$. On an intersected interpolation piece evaluate $V_i + D_i(X - X_i)$, with X restricted to $[X_i^-, X_{i+1}^+]$; omit the left restriction for the first piece and the right restriction for the last piece. Hull the values and the derivatives of the intersected pieces. For the recorded evaluation order, start with the piece preceding the first knot whose upper enclosure endpoint is at least X^- , clamping the piece index to the first or last piece if necessary. Visit pieces in increasing order and stop after piece i when $X_{i+1}^- \geq X^+$. On a nondegenerate input interval include only positive-length intersections, since their limiting values cover the endpoints; for a singleton input include singleton intersections. In particular, at an isolated exactly representable knot use its left piece when it has one, and use the first piece otherwise. A point's derivative is not used in the centered lower bound below. For the cap, use the five-cost evaluation order in Section 5.5, restricting first to the four L^* pieces and including one-sided derivatives at every seam.

The complete parameter checks. Write $x_0 < \dots < x_{N-1}$ for the knots. Check every complete interval $[X_i^-, X_{i+1}^+]$, extending the first left endpoint and the last right endpoint to the outward endpoints of $\mathcal{I}$. On each interval evaluate U, z, h, M and then, in order,

$$\begin{aligned} q &:= b - z^2 - h^2, & \varrho &:= (\beta/b)h, & b_0 &:= \sqrt{3q}, & v &:= \beta + h^2 + e, \\ d_0 &:= v + z^2, & D &:= \sqrt{v(d_0 - 3q)}, & y_- &:= (b_0 z - D)/d_0, & y_+ &:= (b_0 z + D)/d_0. \end{aligned}$$

Require strictly positive lower endpoints for

$$q, \quad h, \quad M - U^2 - h^2, \quad \varrho^2 - h^2 - e, \quad b_0 - \sqrt{z^2},$$

and require $y_-^- > [-5/8]^+$ and $y_+^+ < [5/8]^-$. Over all these intervals record

$$u_- := \min U^-, \quad u_+ := \max U^+, \quad q_- := \min q^-, \quad h_- := \min h^-.$$

Here the interval minima and maxima range over the complete interpolation pieces, not merely over their knots.

For the first control evaluate the same $q, \varrho, b_0, v, d_0, D, y_-, y_+$ with $(z, h) = (z_1, h_1)$. Require strictly positive lower endpoints for $q_1, h_1, \varrho_1^2 - h_1^2 - e$, and $b_{0,1} - \sqrt{z_1^2}$, and the same strict bounds on the retention roots. With $\lambda_0 := \sqrt{3b} - e$ require

$$(\beta + e - \lambda_0^2 - h_1^2)^- > 0.$$

Finally require strictly positive lower endpoints for all of

$$\begin{aligned} u_- - 3/4, \quad 9/10 - u_+, \quad q_-, \quad h_-, \quad \min\{h_-, h_1^-\} - 7/100, \\ e/2, \quad e, \quad \beta + e - 3b, \quad 3/4 - \sqrt{139}/16, \quad 1652275/10^7 - a_0. \end{aligned}$$

In expressions involving an already computed endpoint such as u_- , that endpoint is a singleton interval. The check of $e/2$ supplies both the ordinary and the tilted-projection peeling reserves. Also require $(T_*\beta - (1 - a_0 + e/8))^- > 0$, which verifies that the main target lies strictly inside the specialized local range. Also form

$$J := [[-1/2 - 10^{-10}]^-, [3/4 + 10^{-11}]^+]$$

and require $(1 + 2JJ)^- > 0$, where JJ is the interval product, not the squaring operation. Its two factors represent independent correlations. This verifies corner monotonicity throughout both the main domain and the larger first-parent collar.

The rectangles and their values. Form a sorted list of distinct dyadic cuts from the outward endpoints of $\mathcal{I}$, the two endpoints of $[k/100]$ for $-50 \leq k \leq 75$, and the endpoints of the interval expressions

$$(3L - 1)/(6L + 2), \quad (1 - 3L)/4, \quad L \in \{[1/20], [1/10]\}.$$

The cuts 0 and $1/4$ are included as well. Let C_j be the closed interval between successive cuts. The seed rectangles are $J_0 \times C_j$, where

$$J_0 := [[-1/2 - 10^{-10}]^-, [-1/2 + 10^{-10}]^+].$$

The main rectangles are all $C_i \times C_j$. Each initial rectangle has depth zero. First finish the parameter and seed checks, then check all main rectangles.

For a rectangle $X \times T$, evaluate $y(s, \tau) = (s + 2\tau)/(\sqrt{3}\sqrt{1 - s^2})$ at the singleton interval inputs (X^-, T^-) and (X^+, T^+) , obtaining the intervals y_{low} and y_{high} , respectively. Each evaluation uses the interval operations above in order, not a single rounding of the final value. Prune the rectangle if $X^- + T^- > [1/2 + 10^{-11}]^+$, or if $y_{\text{low}}^- > [5/8]^+$ or $y_{\text{high}}^+ < [-5/8]^-$. If it has not been pruned and either $y_{\text{low}}^- \leq -1$ or $y_{\text{high}}^+ \geq 1$, request subdivision before evaluating the remaining square roots.

Otherwise evaluate the natural interval expression for y and intersect its value interval with $[y_{\text{low}}^-, y_{\text{high}}^+]$. Keep its natural derivative intervals unchanged. On a main rectangle use the parent profile control; on a seed rectangle use the constant first control. Compute

$$r_s := \sqrt{1 - s^2}, \quad r_\tau := \sqrt{1 - \tau^2}, \quad A := (1 + 2s\tau)/(2r_s), \quad b_c := \sqrt{3q} - zy,$$

$$P := b_c^2 - (\beta + e + h^2)(1 - y^2), \quad B := \kappa b_c, \quad d := \kappa(\sqrt{1 - y^2}\sqrt{\beta + \varrho^2} - b_c)/(\varrho + h), \quad R := r_\tau U(\tau).$$

Prune if $P^- > 0$. If $d^- \leq 0$, request subdivision. All other invalid operations cause failure, rather than acceptance.

The whole-rectangle acceptance tests. On a main rectangle put $m := M(s)$, $U := U(s)$ and compute

$$\xi := \sqrt{m - U^2}, \quad E := (1 + \gamma)R/U - A, \quad H' := dU/\xi - E.$$

If $(H')^- \geq 0$, use $f = G_0$; otherwise use $f = G_*$. In both cases use $g = G_{\text{top}}$, with the expressions from Section 5.4. The direct test is $\min\{f^-, g^-\} \geq [10^{-7}]^+$. On a seed rectangle instead use the single expression

$$f_1 := A\lambda_0 + B - d(\sqrt{\beta + e - \lambda_0^2} - h_1) - R$$

and the direct test $f_1^- \geq [1/200000]^+$.

When the direct test fails, compute a centered lower bound for each required expression. If $X = [p/S, q/S]$, let $c_X := \text{trunc}((p + q)/2)/S$, where truncation is toward zero, and put $r_X := \max\{c_X - X^-, X^+ - c_X\}$. Define c_T, r_T in the same way. At $c := (c_X, c_T)$ recompute the value enclosure $f(c)$. For a main rectangle use the branch G_0 or G_* already chosen on the whole rectangle; do not select its branch again at the center. Set

$$L(f) := \max\{f^-, f(c)^- - \text{ru}(r_X \max |D_s f|) - \text{ru}(r_T \max |D_\tau f|)\},$$

where $\max |a, b| := \max\{|a|, |b|\}$ and the derivative intervals are those on the whole rectangle. Accept a main rectangle if $\min\{L(f), L(g)\} \geq [10^{-7}]^+$, and accept a seed rectangle if $L(f_1) \geq [1/200000]^+$. Lemma 5.7 also applies to this rounded center: its proof uses only the two maximum coordinate distances, not equality of those distances on opposite sides.

Subdivision and completion. A rectangle that reaches the residual tests but is not accepted causes failure if its depth is at least 64 or both its widths are at most $8/S$. Otherwise subdivide it. The earlier safety requests for subdivision, made before the residual tests, also subdivide the rectangle. Every subdivision must have a strictly interior dyadic cut; if there is none, the check fails. Thus repeated safety requests cannot silently discard a rectangle.

The recorded execution divides the longer coordinate, choosing s on a tie. In integer endpoint units, put $m := \text{trunc}((p + q)/2)$ and $w := q - p$. Among the two neighboring members of the sorted set of all outward knot endpoints, first consider the least one at least m , then its predecessor. Use the closer strictly interior candidate if its distance from m is strictly less than $\lfloor w/4 \rfloor$; ties favor the candidate considered first. If neither is eligible, use m . Both child rectangles include the cut and have depth increased by one. Process the child with the smaller coordinate first. The initial rectangles are placed on a last-in-first-out list in increasing child-interval index and then increasing parent-interval index; the seed list has only the child index. These ordering conventions specify the recorded subdivision, but they play no role in the validity of an accepted or pruned rectangle.

Success requires every parameter test to pass and the pending lists for both seed and main rectangles to become empty, with every removed rectangle either accepted or pruned by one of the stated rules. An interruption, invalid operation, or unresolved rectangle is not success. Induction under the closed subdivision rule proves complete coverage, and the interval and centered bounds prove the asserted margins on every accepted rectangle.

Certified margins and completion. For the generated rational profile in Definition 5.4, all the parameter, seed, and main checks above pass. An independent complete replay also verifies both required inequalities. The least certified seed and amplified numerator bounds are, respectively,

$$\frac{419729035}{2^{46}} > \frac{1}{200000}, \quad \frac{7037249}{2^{46}} > 10^{-7}.$$

The separately implemented complete check gives $478374871/2^{46}$ and $7037593/2^{46}$, respectively, also above the two required bounds. The seed margin is before division by $r(\tau) \leq 1$; the amplified margin is before division by $r(\tau)U(\tau) < 1$. Every strict parameter test passes on each complete interpolation piece, including both endpoint collars and the separate first control. Every initial and main rectangle is either accepted or pruned by the specified necessary conditions; no unresolved rectangle remains. Thus the closed rectangles cover every retained pair, and their bounds, together with the analytic elimination of the full threshold interval, prove Lemma 5.6. The local checks in Section 5.1 similarly cover every prescribed loss interval.

The complete rational profile, constants, interval operations, and acceptance and subdivision rules are specified in this section.

6 Consequences for 5-, 7-, and 9-colorable graphs

Table 2: Consequences for coloring 5-, 7-, and 9-colorable graphs.

Promise	Authors	Refs	Previous bound	This paper
5-colorable	Halperin, Nathaniel, and Zwick	[HNZ01]	$\tilde{O}(n^{0.468600})$	$\tilde{O}(n^{0.462187})$
7-colorable	Halperin, Nathaniel, and Zwick	[HNZ01]	$\tilde{O}(n^{0.600864})$	$\tilde{O}(n^{0.599584})$
9-colorable	Halperin, Nathaniel, and Zwick	[HNZ01]	$\tilde{O}(n^{0.681630})$	$\tilde{O}(n^{0.681314})$

Halperin, Nathaniel, and Zwick [HNZ01] color k -colorable graphs with $\tilde{O}(n^{\alpha_k})$ colors, where the sequence α_k obeys a recursion whose only external input is an algorithm for 3-colorable graphs. Theorem 4.25 improves that input, and the improvement propagates to every odd k .

We first recall the recursion and prove its monotonicity, then evaluate it for 5-, 7-, and 9-colorable graphs. We finish by explaining why the bounds for even k are unchanged.

Lemma 6.1 (Combined-Color, Theorem 5.1 of [HNZ01]). *Let $0 \leq a < 1$ and suppose there is a randomized polynomial-time algorithm that colors every 3-colorable n -vertex graph with $\tilde{O}(n^a)$ colors. Define*

$$\alpha_2(a) := 0, \quad \alpha_3(a) := a, \quad \alpha_k(a) := 1 - \frac{6}{k+4+3(1-\frac{2}{k})\frac{1}{1-\alpha_{k-2}(a)}} \quad (k \geq 4). \quad (27)$$

Then for every $k \geq 2$ there is a randomized polynomial-time algorithm that colors every k -colorable n -vertex graph with $\tilde{O}(n^{\alpha_k(a)})$ colors.

The algorithm is Combined-Color of [HNZ01], which is stated there with the value $a = 3/14$ of [BK97]. That value enters the analysis only as the exponent of the 3-colorable subroutine: once as the base case $k = 3$, and once through the recursive call on a common neighborhood $N(u) \cap N(v)$, which is the step that contributes α_{k-2} . Any algorithm meeting the stated guarantee may therefore be substituted. As [HNZ01] observes, that recursive step is written for a deterministic subroutine; a randomized one is accommodated by repeating it until the failure probability is negligible, at a polynomial cost.

We check that improving the three-colorable base case propagates monotonically through the recursion.

Lemma 6.2 (Monotonicity). *For every $k \geq 2$ the map $a \mapsto \alpha_k(a)$ is nondecreasing on $[0, 1]$.*

Proof. We induct on k , the cases $k = 2$ and $k = 3$ being immediate from the definition. Let $k \geq 4$, put $K := k + 4$ and $C := 3(1 - \frac{2}{k})$, both positive, and write $\alpha_k(a) = f(\alpha_{k-2}(a))$ with

$$f(x) := 1 - \frac{6}{K + \frac{C}{1-x}}.$$

On $x < 1$ the map $x \mapsto 1 - x$ is positive and decreasing, so $x \mapsto C/(1 - x)$ is increasing, so $x \mapsto K + C/(1 - x)$ is increasing and positive, so $x \mapsto 6/(K + C/(1 - x))$ is decreasing, and therefore f is increasing. Composing f with the inductive hypothesis for $k - 2$ gives the claim. $\square$

Substituting the main exponent into the odd branch gives the following explicit palette bounds.

Theorem 6.3 (Odd palettes). *There are randomized polynomial-time algorithms coloring every 5-, 7-, and 9-colorable n -vertex graph with, respectively,*

$$\tilde{O}(n^{191303/413909}), \quad \tilde{O}(n^{4666615/7783099}), \quad \tilde{O}(n^{17132551/25146367})$$

colors. The exponents are smaller than 0.462187, 0.599584, and 0.681314.

Proof. Apply Lemma 6.1 with $a := 1652275/10^7$, as supplied by Theorem 4.25. For $k = 5, 7, 9$, respectively, Eq. (27) gives

$$\begin{aligned} \alpha_5 &:= 1 - \frac{6}{9 + \frac{9}{5(1-a)}} = \frac{191303}{413909}, \\ \alpha_7 &:= 1 - \frac{6}{11 + \frac{15}{7(1-\alpha_5)}} = \frac{4666615}{7783099}, \\ \alpha_9 &:= 1 - \frac{6}{13 + \frac{7}{3(1-\alpha_7)}} = \frac{17132551}{25146367}. \end{aligned}$$

In each display the first step substitutes the indicated k in the recursion; the second evaluates the resulting rational expression. The decimal values are 0.462186132700..., 0.599583148049..., and 0.681313169413..., proving the upward-rounded bounds. $\square$

The parity of the recursive index determines which bounds depend on the three-colorable base case.

Remark 6.4 (Even palettes are unchanged). The recursion in Eq. (27) splits into two chains that never meet. The even chain bottoms out at $\alpha_2 = 0$ and never consults a , so $\alpha_4 = \frac{7}{19}$, $\alpha_6 = \frac{43}{79}$ and $\alpha_8 = \frac{175}{271}$ are untouched by Theorem 4.25; only odd k improves. By Lemma 6.2, any further improvement of the 3-colorable exponent lowers every odd α_k as well.

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each day. Progress then stalled at 0.167 for several days. After Codex 6 Astra and ChatGPT 6 Pro were released, the author immediately used these tools to break the $1/6$ barrier and achieve the current bound. Getting below $1/6$ with the new models took only one or two prompts. The title of this paper was suggested by Victor Reis. All of the proofs have been carefully verified and rewritten by the author. The author takes full responsibility for the correctness of the proofs in this work.